

Electronic Companion

Limited-Memory Renewal Contracts: Harmonic Coarsening and Certified Joint Design

This companion retains the preceding mathematical chain, the comparison certificates, resource accounting, and computational tables. Complete study narratives and implementation protocols appear in the separately identified Computational Reproduction Record. Inherited protocols and records are identified by their revision directories; they are not relabeled as new executions.

EC.1 The Contract and Its Execution Boundary

EC.1.1 A saturated renewal agreement

At the renewal date, a public branch $j \in \{1, \dots, k\}$ is observed with probability $\pi_j > 0$, where $\sum_j \pi_j = 1$. Its continuation cap is $0 < b_1 < \dots < b_k < 1$. The provider chooses an intermediate tier $Y_j \in [0, 1]$ and a terminal tier $C_j \in [0, 1]$. All branches subsequently reach the same terminal public vertex. Unit payment coefficients and unit discounting make total remaining payment $Y_j + C_j$. A fixed service-benefit continuation minus the customer's outside continuation gives the cap. At the root the provider has committed to

$$\sum_j \pi_j \mathbb{E}[Y_j + C_j] = \bar{b}, \quad \bar{b} := \sum_j \pi_j b_j. \quad (\text{EC.1})$$

Under expected participation, $\mathbb{E}[Y_j + C_j] \leq b_j$ for every j , with expectation conditional on that observed branch and taken before the internal draw. Positivity of the probabilities and (EC.1) imply equality separately on every branch. Under pathwise participation, the stronger constraint is $Y_j + C_j \leq b_j$ almost surely. The same argument then implies equality almost surely, not merely in expectation.

The operating payoff is $f(C_j) - h_j(Y_j)$, with increasing concave f and convex nondecreasing h_j satisfying $h_j(0) = 0$. General shape assumptions suffice for the deterministic and pathwise results. The global continuous algorithm specializes to the quadratic primitives in (54). The negative intermediate contribution represents the net cost of an advance or expedited tier, while the terminal tier contributes net operating return. It does not mean that the customer has negative service benefit: that benefit has already entered the continuation cap. There are no endogenous public transitions, coupled commitments, or switching costs in the renewal theorem.

EC.1.2 What a writable code means

At the observed branch, an encoder selects a symbol $Z \in \{1, \dots, m\}$. At the terminal vertex the decoder observes only the current public vertex and Z . Its stored program can contain arbitrary read-only model constants but cannot recover the realized branch from those constants. An inherited intermediate tier, a customer identifier with a branch lookup, a retained shared random seed, and a transmitted numerical payment are all additional history channels; none is free in this interface. An m -symbol alphabet requires $\lceil \log_2 m \rceil$ fixed-width writable bits, not m bits. Public-state storage and read-only program storage are counted separately.

Private encoder randomization is allowed under expected participation. Fresh decoder randomness is independent of earlier private choices conditional on the symbol. Any seed that correlates those choices across dates must itself be encoded in the charged symbol. Concavity permits a deterministic terminal decoder at the expected-participation optimum, so a codebook is a collection of terminal levels. The lower bound for exact implementation continues to allow randomized encoders and decoders under this accounting. Distinct full-information terminal actions and strict concavity remain visible assumptions; Proposition EC.4.2 handles repeated caps; uniqueness is required for the exact-action lower bound.

EC.1.3 Acceptance timing

The renewal desk observes the branch and offers the intermediate tier and terminal lottery. Under expected participation, the customer accepts before the ticket draw; after acceptance only the ticket reaches the common gateway. The realized ticket may be disclosed, but no new costless exit option is supplied after the draw within this episode. Under pathwise participation, each possible realized obligation must instead satisfy the branch cap, and a draw-specific intermediate tier is allowed. A seed known before acceptance changes that conditioning information. Repeated-customer effects require an enlarged state and are not part of the three-date theorem. The full stipulated mechanism and its qualifications are given in the electronic companion; no empirical or legal claim about an actual provider is made. Section EC.4 allows a nonsaturated root promise, and Section EC.9 specifies the intermediate risk protocol.

EC.1.4 Preferences and a restricted execution interface

A concrete preference interpretation of expected participation is quasilinear, risk-neutral continuation utility $A_j - \mathbb{E}(Y + C)$ with outside value O_j , so that $b_j = A_j - O_j$ after normalization. Alternatively the expectation bound can be an explicitly agreed actuarial budget. Neither interpretation follows from a finite communication alphabet. A customer with a hard realization-level liquidity limit instead requires $Y + C \leq b_j$. The intermediate institution retains the actuarial cap and separately authorizes at most δ realization overrun. Its intermediate tier is committed before the draw so that the disclosed lottery and its ceiling can be checked together. Allowing a fresh, costless rejection after a high draw changes the feasible problem; the expected-participation comparison assumes that option is absent, not that every real renewal contract can eliminate it.

The downstream restriction is an interface restriction, not an inability to store a model. An upstream desk keeps the branch record; a downstream gateway has a read-only map from a small session token to tested execution levels and no access to the upstream record or a branch-indexed customer identifier. A relevant documented component analogy is operational-technology allowlisting and least functionality: NIST SP 800-82 Rev. 3 discusses restricting executable applications, testing policy configurations, disabling nonessential interfaces, and restricting writable set points (Stouffer et al., 2023, Sections 5.2.5.1, 6.2.4.1 and control CM-7). These are precedents for a constrained execution boundary, not evidence that a particular provider uses our entire renewal or lottery protocol.

Under this design, opening an additional level can incur a one-time cost of engineering, test fixtures,

authorization, and configuration assurance. This is the modeled source of $\rho(c)$: it is paid if the level is enabled, independently of the frequency with which its token is used. Per-use charges and entropy-based communication charges would be different objective terms. An exogenous catalog represents levels already eligible for certification; a numerical catalog chosen by the operator instead calls for the approximation theorem in Section E. The model does not assert that every read-only contract must be inaccessible downstream; it quantifies the consequence when that access is deliberately excluded. No empirical adoption, calibration, or enforceability conclusion is inferred from the analogy.

EC.2 The Deterministic Service-Code Frontier

Consider three dates. The root tier is fixed at zero. At the intermediate date a branch j is observed with probability $\pi_j > 0$, where $\sum_j \pi_j = 1$. All branches then enter the same terminal public vertex. Payments and discounting equal one. Intermediate and terminal tiers are in $[0, 1]$. Branch j has cap $0 < b_1 < \dots < b_k < 1$, and the root promise is $\bar{b} = \sum_j \pi_j b_j$. At the intermediate date the reward is $-h_j(y)$, where h_j is convex, nondecreasing, and $h_j(0) = 0$. The common terminal reward $f(c)$ is continuous, concave, and strictly increasing on $[0, 1]$. The terminal cap is one. The terminal decoder observes only its public vertex and its memory symbol, not the inherited intermediate tier.

Theorem EC.2.1 (Heterogeneous renewal frontier). *The unique full-information policy is $y_j = 0$, $c_j = b_j$, with value $J = \sum_j \pi_j f(b_j)$. Exact implementation requires k symbols under strict cap ordering; repeated caps are covered by Proposition EC.4.2. With at most m deterministic terminal symbols, an optimal encoding has contiguous nonempty cells in cap order. For a cell $i, \dots, j$, its terminal tier is b_i and its loss is*

$$C(i, j) = \sum_{h=i}^j \pi_h [f(b_h) - f(b_i) + h_h(b_h - b_i)]. \quad (\text{EC.2})$$

The exact minimum loss is $D_m = D(\min(m, k), k)$, where

$$D(s, j) = \min_{s-1 \leq i < j} \{D(s-1, i) + C(i+1, j)\}, \quad D(0, 0) = 0, \quad (\text{EC.3})$$

and all other infeasible entries equal $+\infty$. With a precomputed cell-cost table this requires $O(mk^2)$ arithmetic operations. For quadratic f and quadratic h_j , prefix sums of the probability-weighted coefficients compute each cell cost in constant arithmetic time.

The proof is in Electronic Companion Section EC.5.1.

This strengthens and includes the quadratic equal-probability frontier. With $f(c) = 2c - c^2/2$, $h_j(y) = y^2/2$, and $\pi_j = 1/k$, (EC.2) becomes

$$C(i, j) = \frac{2 - b_i}{k} \sum_{h=i}^j (b_h - b_i). \quad (\text{EC.4})$$

For $k = 8$ and $b_j = j/9$, one, two, and three symbols lose $119/162$, $5/18$, and $49/324$, respectively. Two

writable bits permit four symbols, not three. The segmentation recurrence is standard; the contract-specific cell loss, ordered-cell sufficiency, and exact memory interpretation are the substantive statements.

EC.3 When Randomization Improves a Restricted Memory Budget

The exact-optimum lower bound in Theorem EC.4.1 does not imply that randomization is useless below that alphabet size. The renewal family permits an explicit answer under the declared conditional-expectation participation institution. At an intermediate public branch, the controller may randomize its terminal symbol; it carries no uncharged seed or branch record. Participation is evaluated before that private draw. Terminal actions may depend only on the terminal public vertex and the retained symbol.

For terminal levels $c_1 < \dots < c_s$, let F_c be the linear interpolant through $(c_i, f(c_i))$. Since f is concave and strictly increasing, this interpolant is concave and strictly increasing on $[c_1, c_s]$; for $s = 1$ it is defined at its sole point. Retain the caps, probabilities, and convex intermediate costs of Theorem EC.2.1.

Theorem EC.3.1 (Optimal lotteries for a fixed terminal alphabet). *For a fixed set of terminal levels, a feasible randomized controller exists if and only if $c_1 \leq b_1$. Its optimum at branch j has deterministic intermediate tier and terminal mean*

$$\mu_j = \min\{b_j, c_s\}, \quad y_j = b_j - \mu_j.$$

It randomizes over at most the two adjacent terminal levels bracketing μ_j . The optimal loss with at most m writable terminal symbols is consequently

$$D_m^{\text{ran}} = \min_{\substack{1 \leq s \leq m, \\ 0 \leq c_1 < \dots < c_s \leq 1 \\ c_1 \leq b_1}} \sum_j \pi_j \{f(b_j) - F_c(\min\{b_j, c_s\}) + h_j((b_j - c_s)_+)\}. \quad (\text{EC.5})$$

The minimum allows coincident levels by deleting unused duplicates. It satisfies $D_m^{\text{ran}} \leq D_m$. Equality with the full-information optimum still requires k symbols when the full-information actions are distinct and rewards are strictly concave. Formula (EC.5) optimizes the encoding for each continuous codebook; it is not a claim that the joint codebook problem is solved by the deterministic segmentation recurrence.

The proof is in Electronic Companion Section EC.5.2.

Proposition EC.3.2 (A strict, exactly solved two-symbol separation). *In the quadratic renewal model, take three equiprobable branches with caps $1/4, 1/2, 3/4$, $f(c) = 2c - c^2/2$, and $h_j(y) = y^2/2$. Then*

$$D_2 = \frac{1}{8}, \quad D_2^{\text{ran}} = \frac{1}{96}, \quad D_3 = D_3^{\text{ran}} = 0. \quad (\text{EC.6})$$

Two symbols thus reduce the randomized loss to one twelfth of the best deterministic loss, without reducing the three-symbol requirement for exact implementation.

The proof is in Electronic Companion Section EC.5.3.

The institutional qualification is substantive. On the middle branch, the high private draw delivers terminal payment $3/4$ against an expected-continuation cap $1/2$. The lottery is feasible when participation is

checked before the private draw, as stipulated here, but not when that branch cap must hold for every private draw. We do not transfer the randomized frontier to the pathwise institution. The deterministic frontier and the exact-memory lower bound retain their original domains. This separation explains why a statement about the minimum alphabet at the exact optimum cannot settle the economic value of randomization under a smaller alphabet.

EC.4 Promise Allocation Away from Saturation

The saturated promise is an endpoint of a larger allocation problem, not a normalization. Fix the branch caps and replace the root promise by $B \in [0, \bar{b}]$, where $\bar{b} = \sum_j \pi_j b_j$. Write $V_m^E(B)$, $V_m^P(B)$, and $V_m^D(B)$ for optimal value with at most m symbols under expected participation, pathwise participation allowing draw-specific intermediate tiers, and deterministic execution, respectively. The information interface remains unchanged. There is no codeword-dependent charge in this section, and terminal levels may range continuously over $[0, 1]$. The full-information value is $V_\infty(B)$.

EC.4.1 An exact threshold for every root promise

Theorem EC.4.1 (Clipped service levels and exact alphabet). *Suppose f is strictly increasing and strictly concave, and each h_j is convex, nondecreasing, with $h_j(0) = 0$. For every $B \in [0, \bar{b}]$, full information uniquely implements*

$$Y_j = 0, \quad C_j = \min\{b_j, \tau\}, \quad \sum_j \pi_j \min\{b_j, \tau\} = B. \quad (\text{EC.7})$$

At $B = 0$ take $\tau = 0$, and at $B = \bar{b}$ take $\tau = \max_j b_j$. The minimum alphabet attaining this value under any of the three institutions is

$$n(B) = |\{\min(b_j, \tau) : 1 \leq j \leq k\}|.$$

In particular, for $0 \leq B \leq \min_j b_j$, one symbol attains $V_m^I(B) = f(B)$ for every $m \geq 1$ and $I \in \{E, P, D\}$.

Proof. For a full-information feasible policy, let $t_j = \mathbb{E}(Y_j + C_j) \leq b_j$. Replacing it on branch j by $Y_j = 0, C_j = t_j$ preserves the root promise and satisfies even pathwise participation. Jensen's inequality, strict increase of f , and $h_j \geq 0$ show that the replacement weakly improves value, strictly if $\mathbb{E}Y_j > 0$ or C_j is nonconstant. Hence it suffices to maximize $\sum_j \pi_j f(t_j)$ with $0 \leq t_j \leq b_j$ and $\sum_j \pi_j t_j = B$. Strict concavity implies uniqueness. If two uncapped coordinates differ, transferring equal probability-weighted amounts from the larger to the smaller strictly improves this sum; the same transfer shows that a capped coordinate below the common free level must be filled. Thus the unique optimizer is (EC.7), with the endpoint cases interpreted as stated. Continuity of the clipped mean gives existence of τ .

Equality in these improvements requires deterministic terminal action $\min(b_j, \tau)$ on each branch. A symbol's terminal decoder cannot implement two different deterministic actions at the same public vertex. Its conditional output distribution is independent of the originating branch, including when fresh decoder randomization is permitted. Consequently distinct optimal actions require distinct symbols. Assigning one

symbol per distinct action attains the lower bound under all institutions. When $B \leq \min b_j$, all optimal actions equal B . $\square$

For strictly ordered caps, the thresholds at which $n(B)$ changes are explicit: put $B_\ell = \sum_j \pi_j \min(b_j, b_\ell)$. The exact alphabet is $\ell + 1$ for $B_\ell < B \leq B_{\ell+1}$, $1 \leq \ell < k$, and it is one on $[0, B_1]$. At $B = B_\ell$ it is ℓ . Thus exact implementation has a promise-dependent phase diagram, rather than requiring k symbols at every promise.

Proposition EC.4.2 (Repeated caps at the saturated endpoint). *For $B = \bar{b}$, branches with a common cap b can be aggregated in every deterministic, expected, and pathwise frontier. Their aggregate probability is $p_b = \sum_{j:b_j=b} \pi_j$ and their aggregate intermediate cost is $h_b(y) = p_b^{-1} \sum_{j:b_j=b} \pi_j h_j(y)$. Quadratic costs therefore aggregate by their probability-weighted curvature. The exact alphabet threshold is the number of distinct caps. For arbitrary B , the full-information threshold remains (EC.7), but aggregation of the restricted-alphabet problem with heterogeneous intermediate costs is not asserted.*

Proof. At saturation, the fixed-book expected rule chooses the same terminal mean and intermediate tier for all branches with the same cap. The deterministic and pathwise rules choose the same highest eligible terminal level. Summing their rewards gives precisely the aggregate cost. Optimize over the common books and apply Theorem EC.4.1 for the threshold. Away from saturation the branch promises themselves can differ, so the saturated aggregation argument does not apply. $\square$

EC.4.2 Uniform transport between any two root promises

Assume in addition that f is Lipschitz with constant L_f on $[0, 1]$ and each h_j is Lipschitz with constant H_j on $[0, b_j]$; put $H = \max_j H_j$. These constants are independent of the alphabet budget. For quadratic primitives one may take $L_f = r$ and $H = \max_j \gamma_j b_j$. The smaller intermediate domain is sufficient because the mean and pathwise replacements used below have tiers at most b_j .

Theorem EC.4.3 (All-promise value and institutional-gap stability). *For any $0 \leq B_1 \leq B_2 \leq \bar{b}$, $\Delta = B_2 - B_1$, every $m \geq 1$, and $I \in \{E, P, D\}$,*

$$V_m^I(B_2) - L_f \Delta \leq V_m^I(B_1) \leq V_m^I(B_2) + H \Delta. \quad (\text{EC.8})$$

If $G_m(B) = V_m^E(B) - V_m^P(B)$, then

$$|G_m(B_2) - G_m(B_1)| \leq (L_f + H) \Delta. \quad (\text{EC.9})$$

The same bound holds for changes in the loss $V_\infty(B) - V_m^I(B)$. In particular, a certified strict advantage $G_m(B_0) > 0$ persists for every feasible B with $|B - B_0| < G_m(B_0)/(L_f + H)$.

Proof. First contract any policy at $B_2 > 0$ by multiplying all intermediate and terminal actions by $\alpha = B_1/B_2$. The decoder uses the same symbols, and both participation conventions remain feasible. Intermediate cost weakly decreases. The decrease in terminal reward is at most $L_f(1 - \alpha)\mathbb{E}C \leq L_f \Delta$, where expectations include branch weights. The case $B_2 = 0$ is immediate. Taking suprema gives the left inequality.

For the other direction, start from a policy at B_1 . Under expected participation, replace the terminal decoder given a symbol by its mean and the intermediate tier on each branch by its mean. These changes weakly improve reward, preserve the alphabet, and preserve expected obligations. Write $\mu_j = \mathbb{E}C_j$, $y_j = \mathbb{E}Y_j$, and $s_j = b_j - y_j - \mu_j \geq 0$. If $B_1 < \bar{b}$, put $\theta = \Delta/(\bar{b} - B_1)$ and replace y_j by $y_j + \theta s_j$. The total added obligation is $\theta \sum_j \pi_j s_j = \Delta$. Its cost is at most $H\Delta$, and the book and its encoding probabilities are unchanged.

For pathwise participation, replace fresh terminal decoder randomness conditional on symbol z by its mean c_z . Conditional independence from the earlier tier ensures $Y_j + c_z \leq b_j$ almost surely after this replacement; terminal concavity weakly improves reward. Use the draw-specific tier

$$Y'_j = Y_j + \theta(b_j - Y_j - c_z).$$

This is admissible at the stipulated intermediate timing, remains between Y_j and $b_j - c_z$, and adds expected obligation Δ with cost at most $H\Delta$. It requires no extra information at the terminal gateway. The deterministic case is the same construction without random draws. If $B_1 = \bar{b}$, necessarily $\Delta = 0$. Taking suprema, or limits of near-optimal policies, proves the right inequality without requiring a maximizer.

For each institution the value increment lies in $[-H\Delta, L_f\Delta]$. Subtracting two such increments proves the institutional-gap bound. The same constructions apply with unrestricted full information, yielding the loss bound. Every transformation preserves the original number of writable symbols; no concavity of the optimized codebook envelope is assumed. $\square$

In the equal-probability three-branch example of Proposition EC.3.2, $\bar{b} = 1/2$, $G_2(\bar{b}) = 11/96$, and $L_f + H = 2 + 3/4 = 11/4$. Therefore

$$G_2(B) \geq \frac{11}{96} - \frac{11}{4} \left(\frac{1}{2} - B \right) > 0 \quad \text{for } \frac{11}{24} < B \leq \frac{1}{2}. \quad (\text{EC.10})$$

This is a sufficient certified interval of strict advantage, not a maximal interval or an exact nonsaturated loss formula. The global cell algorithm below computes exact values outside this conservative interval as well. At the other endpoint, $B \leq 1/4$ gives zero loss with one symbol. In between, cap anchoring at the original b_j and pathwise–deterministic equality do not follow from the saturated proofs. Indeed the one-symbol full optimum $C = B < \min b_j$ already shows why original-cap anchoring cannot hold for all promises. The decision insight persists without implying that the saturated algorithm solves every interior promise.

EC.4.3 The nonsaturated allocation problem

The larger problem has an exact reduction even when it does not have the saturated finite-anchor reduction. For a fixed book $c = (c_1, \dots, c_s)$ let F_c be its concave interpolant and define

$$v_{j,c}(t) = F_c(\min\{t, c_s\}) - h_j((t - c_s)_+), \quad c_1 \leq t \leq b_j. \quad (\text{EC.11})$$

Proposition EC.4.4 (Joint promise allocation and codebook design). *The expected-participation value at*

any B is exactly

$$V_m^E(B) = \max_{\substack{|c| \leq m, 0 \leq c_1 < \dots < c_s \leq 1 \\ c_1 \leq t_j \leq b_j, \sum_j \pi_j t_j = B}} \sum_j \pi_j v_{j,c}(t_j). \quad (\text{EC.12})$$

For a fixed book this is a separable concave resource-allocation problem. Branch targets are the chosen t_j , not mechanically the original caps. Saturation fixes $t_j = b_j$ and recovers the earlier frontier.

Proof. Set $t_j = \mathbb{E}(Y_j + C_j)$ for an arbitrary feasible policy. The mean replacements in Theorem EC.3.1 are still valid. For a given t_j , maximizing $F_c(\mu) - h_j(t_j - \mu)$ over $c_1 \leq \mu \leq \min(t_j, c_s)$ selects its largest feasible mean by monotonicity. Conversely that mean and its adjacent lottery implement (EC.11) with intermediate tier $(t_j - c_s)_+$. This proves both directions of (EC.12). The response is concave: to the left of c_s it is F_c , to the right it is $f(c_s) - h_j(t - c_s)$, and the one-sided slopes drop from a nonnegative chord slope to a nonpositive cost slope. This also covers a singleton book. Allocation with a common supporting multiplier, allowing subgradient intervals at kinks, solves the fixed-book problem. No interchange of codebook optimization and a dual minimization is used. $\square$

Electronic Companion Proposition EC.8.1 gives an exact supporting-price algorithm and a separately checkable certificate for every fixed-book allocation. Exhaustive certified catalog search extends that computation to all promises on small catalogs; its exponential cost is stated explicitly.

Theorem 2.1 extends the fixed-book response to every realization ceiling. Theorems B.2 and E.1 solve the continuous outer design and certify its catalog approximation; thus the inner allocation is no longer the endpoint of the nonsaturated analysis.

EC.5 Proofs for the Saturated Continuous Frontier

EC.5.1 Proof of Theorem EC.2.1

Proof. The root promise is the mean of the branch caps, so every branch cap must bind. Hence $y_j = b_j - c_j$ and $0 \leq c_j \leq b_j$. The branch reward $f(c) - h_j(b_j - c)$ is strictly increasing on that interval. Full information therefore uniquely selects $c_j = b_j$; different terminal actions require different symbols.

Within a memory cell all terminal actions coincide and cannot exceed its smallest cap. Monotonicity selects that cap and subtraction from the full-information reward gives (EC.2). Fix any collection of cell minima. Every branch strictly prefers the largest eligible selected minimum, irrespective of its probability and intermediate cost. Reassigning branches in this way keeps each selected minimum in its own nonempty cell and produces contiguous cells without reducing value. Splitting at the final cell gives (EC.3). Expanding the quadratic terms gives the stated prefix-sum implementation. $\square$

EC.5.2 Proof of Theorem EC.3.1

Proof. The root promise equals the weighted mean of the branch caps. Every branch cap must therefore bind in conditional expectation, so $\mathbb{E}(Y_j + C_j) = b_j$. Replace a randomized terminal decoder, conditional on a symbol, by its mean. It uses the same number of symbols, preserves payments, and weakly increases reward

by concavity. Replace a randomized intermediate tier by its branch mean; convexity of h_j weakly lowers its cost without altering the encoder's terminal-symbol probabilities. These replacements use exogenous transitions and the absence of switching costs.

For a terminal mean μ in the convex hull of the levels, the largest expected terminal reward is $F_c(\mu)$. The upper hull of the concave points $(c_i, f(c_i))$ consists of adjacent chords; a lottery over their endpoints attains the hull. The branch objective is $F_c(\mu) - h_j(b_j - \mu)$ on $[c_1, \min(b_j, c_s)]$. It is strictly increasing, so its maximum occurs at μ_j . Feasibility requires $c_1 \leq b_j$ for every branch, equivalently $c_1 \leq b_1$. Subtraction from full information gives the formula. Compactness of symbol probabilities and levels gives attainment, with redundant symbols deleted. Deterministic encodings are admissible randomized encodings, yielding the inequality. The exact-optimum statement follows from the strict-concavity argument in Theorem EC.4.1, not from the restricted-alphabet formula alone. $\square$

EC.5.3 Proof of Proposition EC.3.2

Proof. The two deterministic contiguous partitions have losses $7/48$ and $1/8$. For the randomized problem, let the two levels be $u \leq v$. Any choice with $v < 1/4$ can be improved by raising its levels. For $v \geq 1/4$, feasibility requires $u \leq 1/4$; raising u to $1/4$ weakly increases every relevant concave chord value. Thus take $u = 1/4$.

For v in $[1/4, 1/2]$, $[1/2, 3/4]$, and $[3/4, 1]$, respectively, Theorem EC.3.1 gives loss

$$\begin{cases} [(1/2 - v)(2 - v) + (3/4 - v)(2 - v)]/3, \\ [(v - 1/2)/8 + (3/4 - v)(2 - v)]/3, \\ [(v - 1/2)/4 + (v - 3/4)/2]/6. \end{cases}$$

The first two expressions strictly decrease on their domains and the third strictly increases. Hence the global minimum is at $v = 3/4$ and equals $1/96$. The first and third branches use their respective endpoint symbols; the middle branch uses each with probability one half. All intermediate tiers are zero. The middle branch's loss is the strict-concavity loss $1/32$, weighted by its probability $1/3$. Three distinct terminal symbols attain full information. $\square$

EC.5.4 Proof of Lemma C.1

Proof. Attainment follows from the compact level-and-probability formulation in Theorem EC.3.1. Delete unused levels and coincident duplicates. If all levels lie below b_1 , raising the largest to b_1 improves every branch and leaves only that level relevant. Otherwise delete levels below b_1 except the largest such level, if any, and raise it to b_1 . Every affected cap is bracketed by this level and the next one. Equation (55) shows that raising the lower endpoint weakly lowers loss. This also covers a one-level codebook. The same argument at the other extreme deletes superfluous levels above b_k and lowers the remaining highest endpoint to b_k without increasing loss.

Fix the highest level. Take a nonhighest codeword not equal to a cap. Between the nearest caps and its

neighboring codewords, the set of branch caps in each adjacent interpolation interval is fixed. Equation (55) makes the sum of their losses affine in the moving codeword; all other losses are unchanged. Move it to an endpoint that does not increase this affine function. At a cap it becomes anchored. At a collision it can be deleted: no strictly intervening branch cap has been crossed, and the limiting interpolation losses agree. Each move reduces the number of unanchored nonhighest levels or the number of levels. Finitely many moves produce the stated representation. Branches above the fixed highest level are unaffected throughout. $\square$

EC.5.5 Proof of Theorem C.2

Proof. By Lemma C.1, a nontrivial optimal codebook consists of an anchored prefix and one final level. The prefix contributes exactly the sum of its edge costs. Splitting at the preceding anchor proves (57) by induction on the number of anchors. If two successive edges share an endpoint cap, its loss is zero, so no term is double counted. Conditional on the final anchor b_j , every possible highest level belongs to one of the intervals in (58). Differentiating that quadratic gives (59); its quadratic coefficient is strictly positive whenever $\ell < k$ because the remaining probabilities and q are positive. Formula (60) therefore optimizes over the entire remaining continuous domain.

Every term in (61) reconstructs a feasible controller with at most m codewords. A coincident final level is deleted. Conversely, the representation lemma maps an optimal controller into one of those terms, including the one-symbol alternative. The two inequalities prove equality, not merely an upper bound or a stationary-point condition. Recover the final level attaining T_j , backtrack the minimizing indices in P , and apply Theorem EC.3.1 to obtain the encoder probabilities and intermediate tiers.

Prefix sums of π_h , $\pi_h b_h$, $\pi_h b_h^2$, $\pi_h \gamma_h$, $\pi_h \gamma_h b_h$, and $\pi_h \gamma_h b_h^2$ evaluate each edge and each quadratic tail in constant arithmetic work. All T_i require $O(k^2)$ work. The prefix dynamic program requires $O(mk^2)$ work and $O(mk)$ backpointers; edge costs are evaluated on demand rather than stored in a quadratic array. The six moment arrays and all terminal optima require $O(k)$ storage.

For completeness, each moment, edge coefficient, and tail coefficient is a sum of bounded-degree monomials in rational inputs. A conceptual common denominator is a fixed power of the product of input denominators; it has polynomial bit length and is not constructed by the implementation. Each terminal minimizer adds one division of such coefficients. Each prefix cost adds at most k edges; each final candidate adds one tail optimum. Reducing after every operation gives the conservative height bound S . Exact comparisons use cross multiplication on integers of $O(S)$ bits, and ordinary integer arithmetic with greatest-common-divisor reduction costs at most $O(S^3)$ per rational operation. This establishes the stated bound, including comparisons rather than only arithmetic counts. The electronic companion gives an implementation-level concordance. $\square$

EC.5.6 Proof of Proposition C.3

Proof. The representation lemma fixes the lower level at $1/4$. For an upper level $z \in [1/2, 3/4]$, the loss is

$$\frac{3}{40}(z - 1/2) + \frac{1}{20}(3/4 - z)(2 - z) = \frac{z^2}{20} - \frac{z}{16} + \frac{3}{80}.$$

Its unique minimum is at $z = 5/8$ and equals $23/1280$. On $[1/4, 1/2]$, both higher-cap branches lie above z and the loss strictly decreases up to $1/2$. Levels above $3/4$ can be lowered, as in Lemma C.1. These observations also exclude the one-level boundary. The cap-only choices with two feasible levels have losses $3/160$ and $3/160$; one level has loss $49/160$. The deterministic partition using levels $1/4, 1/2$ has loss $3/160$, while the other contiguous two-cell partition has loss $21/80$. Theorem EC.2.1 yields the deterministic optimum.

The middle branch draws terminal level $5/8$ with probability $2/3$ and level $1/4$ with probability $1/3$, with zero intermediate tier. The highest-cap branch uses intermediate tier $1/8$ and terminal tier $5/8$. All branch payments equal their caps in expectation, and the root payment is $17/40$. The resulting weighted losses are $3/320$ and $11/1280$, respectively, summing to $23/1280$. $\square$

EC.5.7 Proof of Theorem C.4

Proof. Pathwise feasibility gives $Y_j + C_j \leq b_j$ almost surely at each branch. The root promise is $\sum_j \pi_j b_j$ and every branch has positive probability. Hence $Y_j + C_j = b_j$ almost surely on every branch: any positive expected slack would contradict root saturation.

A retained symbol determines a terminal level, after replacing an independent randomized terminal decoder by its mean. This replacement preserves feasibility and weakly improves reward. To see this under pathwise saturation, conditional on a branch and symbol the only permissible additional terminal randomization is fresh and independent of the earlier tier; otherwise a retained shared seed would be an uncharged state. Independence together with $Y_j + C_j = b_j$ almost surely forces that decoder to be degenerate. This also shows why the memory architecture matters.

For a fixed codebook, a symbol with terminal level c can be used on branch j only if $c \leq b_j$; saturation fixes its intermediate tier at $b_j - c$. Its payoff is $f(c) - h_j(b_j - c)$, which is strictly increasing in c . Thus the highest eligible codeword weakly dominates every lottery over eligible symbols, including lotteries with draw-specific intermediate tiers. Selecting it deterministically preserves each branch's payment and cannot reduce payoff. Optimizing codebooks yields the deterministic ordered frontier of Theorem EC.2.1. The reverse inequality follows because every deterministic feasible controller is an admissible pathwise-randomized controller. Strict-concavity uniqueness gives the final threshold assertion. $\square$

EC.5.8 Proof of Lemma D.1

Proof. For E , expansion of (56) gives the right-minus-left difference

$$\begin{aligned} \frac{q}{2} \left[(b_d - b_c) \sum_{h=a}^{b-1} \pi_h (b_h - b_a) + (b_b - b_a)(b_d - b_c) \sum_{h=b}^c \pi_h \right. \\ \left. + (b_b - b_a) \sum_{h=c+1}^d \pi_h (b_d - b_h) \right] \geq 0. \end{aligned}$$

Every term is nonnegative, including coincident-index and empty-sum cases. For C , all terms through c cancel and the difference is

$$\sum_{h=c+1}^d \pi_h \{f(b_b) - f(b_a) + h_h(b_h - b_a) - h_h(b_h - b_b)\} \geq 0.$$

Only monotonicity of f and h_h is needed for the second inequality.

Join the functions $\Psi_{i\ell}$ to a continuous function Ψ_i on $[b_i, b_k]$. For $i < j$ and $z \geq b_j$, cancellation gives

$$\begin{aligned} \Psi_i(z) - \Psi_j(z) = \frac{q}{2} & \left[\sum_{h=i+1}^j \pi_h (b_h - b_i)(z - b_h) \right. \\ & \left. + (b_j - b_i) \sum_{b_j < b_h \leq z} \pi_h (z - b_h) \right]. \end{aligned} \quad (\text{EC.13})$$

The expression is continuous and nondecreasing in z : between caps its slope is nonnegative, and every entering summand is zero at its entry cap. Intermediate-curvature terms cancel. Thus $\Psi_i(x) + \Psi_j(y) \leq \Psi_i(y) + \Psi_j(x)$ for $b_j \leq x \leq y$.

For feasible columns $\ell < t$ with $j \leq \ell$, let x minimize Ψ_j on I_ℓ and y minimize Ψ_i on I_t . Compactness ensures attainment and interval order gives $x \leq y$, including a common endpoint. Therefore

$$Q(i, \ell) + Q(j, t) \leq \Psi_i(x) + \Psi_j(y) \leq \Psi_i(y) + \Psi_j(x) = Q(i, t) + Q(j, \ell).$$

The singleton final interval is evaluated directly. This proves the inequality after continuous optimization; convexity of each separate minimization would not by itself suffice. $\square$

EC.5.9 Proof of Lemma D.2

Proof. The two column terms occur once on each side of (64) and cancel. For rows $j < j'$ and columns $i < i'$ with $i' < j$, apply Lemma D.1 to $i < i' < j < j'$ for E , and $i+1 < i'+1 \leq j < j'$ for C . Transposition of the cost arguments gives the required inequality. Unreachable predecessor values are removed, not treated as finite offsets. $\square$

EC.5.10 Proof of Lemma D.3

Proof. Fix $r < s$ and $c < d$ satisfying the antecedent. There are three possible upper-row configurations.

If both entries are feasible, the strict preference means $A(r, d) < A(r, c)$; equality would select c . If both remain feasible in row s , Monge gives $A(s, d) - A(s, c) \leq A(r, d) - A(r, c) < 0$. If c becomes infeasible but d remains feasible, d wins by its zero first key coordinate. If both become infeasible, $-d < -c$ makes d win. A feasible c with infeasible d is impossible for a suffix.

If c is infeasible and d feasible in row r , the nondecreasing cutoff keeps c infeasible in row s . Either d remains feasible and wins by its first coordinate, or both are padded and $-d < -c$ wins. Finally, if both are infeasible in row r , both remain infeasible in row s and the same padding order applies. These cases are

exhaustive. In an entirely infeasible selected row, all keys are $(1, 0, -c)$, uniquely minimized at the largest selected column. Thus rows becoming entirely infeasible after column restriction are included, not excluded by an implicit feasibility assumption. $\square$

EC.5.11 Proof of Lemma D.4

Proof. If $d > u_r$, either c is feasible and wins by its first key coordinate, or both are padded and $c < d$ wins. The antecedent is false in either case. If the antecedent is true, both upper-row entries are finite. Since $u_s \geq u_r$, both remain finite in row s . A finite tie cannot favor d in row r , and the strict finite inequality propagates by Monge exactly as above. On a wholly padded selected row, $(1, 0, c)$ is uniquely minimized at its smallest column. This covers finite ties, feasible/padded comparisons, and padded/padded comparisons without assigning a numerical infinity. $\square$

EC.5.12 Proof of Lemma D.5

Proof. An anchored path with s distinct indices ending at j requires $j \geq s$. Conversely, $1, 2, \dots, s-1, j$ gives a finite path whenever $j \geq s \geq 2$. The one-anchor condition fixes $j = 1$. A deterministic partition of j branches into s nonempty ordered cells exists exactly when $j \geq s$. These are necessary and sufficient characterizations. Discarding endpoint k , which cannot extend to a later endpoint, preserves interval structure. $\square$

EC.5.13 Proof of Theorem D.6

Proof. The finite-cost, offset, completion, and reachability lemmas verify all hypotheses of classical SMAWK (Aggarwal et al., 1987). Each interface call uses $O(k)$ key comparisons. A request either compares padded indices or performs constantly many moment operations and exact cost evaluations, including repeated requests without caching. The six moment arrays cost $O(k)$ work and space. The terminal pass and at most m program layers cost $O(mk)$. Winner reevaluation and last-anchor scans add $O(mk)$ rather than an uncounted quadratic search.

There are $O(mk)$ predecessors, two active value rows, and $O(k)$ extra search indices. Backtracking one winner for each budget writes at most $\sum_{s=1}^m s = O(m^2) \subseteq O(mk)$ symbols. Reconstruction and output are thus included.

Search introduces no new algebraic depth. A terminal call forms the same six moments, coefficient ratio, clamped level, and substituted quadratic value as Theorem C.2. A program call adds one edge or cell to a prior path, and a selected path has at most k such costs. Comparisons evaluate a subset, possibly with repetition, of the candidates of quadratic enumeration. Column offsets change no formula; padding uses indices only. The common-denominator argument of that theorem therefore still bounds evaluated rationals and lottery probabilities by $O(S)$ bits. Reduced-fraction arithmetic and cross multiplication cost conservatively $O(S^3)$ per operation. Table EC.5 records the full accounting. $\square$

EC.6 Detailed Renewal Protocol

Consider a stipulated service-batch agreement administered by a renewal desk and executed by a shared dispatch gateway. The desk observes the customer's current outside alternative and approved continuation. It can perform an advance tier and issue one of a small number of dispatch tickets. The gateway implements the terminal tier named by that ticket; its interface does not retain the renewal record. A separate billing audit can retain the contract without making that record an input to the gateway. Limiting tickets can price gateway-state certification, transmission, or a restricted instruction set. This is a stipulated stylized mechanism, not a claim that a particular industry already uses this architecture.

In the expected-participation version, the offer communicates the advance tier, the terminal ticket levels, and their probabilities. The customer evaluates expected remaining payment before the ticket is drawn and either renews or exits at that review. The protocol commits to the advertised draw. After renewal the desk performs the advance tier, draws the ticket, and sends only that ticket to the gateway. The realized ticket may be disclosed to the customer. "Private" means that the seed is neither a pre-acceptance conditioning variable nor a free future decoder state; it need not mean that realized service is hidden forever. The customer does not receive a new costless exit option after observing an unfavorable draw within this modeled renewal episode.

This version is meaningful only for an ex ante willingness-to-pay or expected-obligation cap and an agreement that permits the stated realization variation. It is not a way to satisfy an ex post enforceable payment ceiling in expectation. A hard billing limit, a service-safety requirement on every draw, or a review after disclosure instead calls for pathwise feasibility, with the deterministic frontier as its optimum. We make no legal assertion about enforceability. The two feasible sets are explicitly different design alternatives.

Repeated customers who may condition a later renewal on past tickets require that history to enter the later public state, cap, or charged memory. The three-date theorem is not an unqualified repeated-game result. Similarly, a contractible seed disclosed before acceptance changes the conditioning information and hence the participation test. It cannot simply be added as a free source of branch-dependent execution. Section EC.15 gives a bounded-realization variant for an operator who permits some variation but not the full expected-participation lottery.

EC.7 Proofs for the Bounded-Overrun Institution

The results in this section use the pre-draw intermediate-action protocol of (EC.15). They do not assert that an arbitrary draw-dependent intermediate-action model reduces to this protocol. All terminal levels and intermediate actions are in $[0, 1]$.

Proposition EC.7.1 (Exact fixed-codebook bounded-overrun rule). *Let f be increasing concave and h_j convex nondecreasing, as in Theorem EC.2.1. Fix a sorted codebook with lowest level at most b_1 . On branch j , let u be the largest codeword at most b_j . If $u = b_j$, use u deterministically. Otherwise, if there is a next codeword $v > b_j$ and $v - b_j \leq \delta$, an optimal choice is the adjacent-code lottery with mean b_j , with $Y_j = 0$. In every other case an optimal choice is the deterministic codeword u , with $Y_j = b_j - u$.*

Proof. Put $\mu = \mathbb{E}C_j$. The committed intermediate action is $Y_j = b_j - \mu \geq 0$, and every codeword used with

positive probability must satisfy $c \leq \mu + \delta$. The greatest expected terminal payoff at a fixed mean is the concave polygonal interpolation $F_c(\mu)$ through the codebook. The function

$$G_j(\mu) = F_c(\mu) - h_j(b_j - \mu)$$

is nondecreasing, and strictly increasing when f is strictly increasing. If $u = b_j$, the deterministic action attains the maximal possible mean. If the top codeword is at most b_j , it similarly attains the maximal possible mean and is optimal.

For the remaining case, $u < b_j < v$. If $v - b_j \leq \delta$, the adjacent lottery with mean b_j satisfies the support bound: its high realization is at most $b_j + \delta$, and its low realization is smaller. It attains both the largest allowable mean and $F_c(b_j)$, and hence the upper bound $G_j(b_j)$.

If $v - b_j > \delta$, no feasible mean can exceed u . Such a mean would require positive mass at a codeword at least v , whereas the realization requirement would imply $v \leq \mu + \delta \leq b_j + \delta$, a contradiction. The deterministic action at u is feasible, attains $G_j(u)$, and weakly dominates every feasible smaller mean. The argument does not permit a tiny mass on an inadmissible high codeword: an almost-sure constraint excludes any positive mass there. $\square$

Proposition EC.7.2 (Global two-code bounded-overrun example). *For the probabilities, caps, and quadratic rewards in Proposition C.3, the minimum loss over all codebooks with at most two levels under (EC.15) is (EC.20). In particular, the optimal highest level is continuous in δ on $[0, 1/8]$ and is off-cap for every $\delta > 0$.*

Proof. We first justify restrictions on the two codewords; they are not imposed by the proposed answer. Feasibility requires the lowest codeword to be at most $b_1 = 1/4$. If both codewords are below $1/4$, increasing the highest to $1/4$ improves every branch's deterministic maximal mean without violating a realization bound. Otherwise write the two levels as $u \leq 1/4 \leq z$. Raising u to $1/4$ cannot worsen an optimal branch payoff. For a feasible mean $\mu \geq 1/4$, use the new adjacent mixture with the same mean: its high endpoint z and the overrun condition are unchanged, while the concave quadratic interpolation increases as the low endpoint rises. A feasible mean below $1/4$ is dominated by deterministic service $1/4$, which all branch caps permit. Thus an optimum has lowest codeword $1/4$, including a one-code solution as a repeated level.

A highest level above $b_3 = 3/4$ can be reduced to $3/4$. For any branch mean at most $3/4$, the adjacent interpolation improves as the upper endpoint falls, and the maximum realization decreases. Conversely, if $z < 1/2$, both the second and third branches are at the top codeword and improving z to $1/2$ strictly raises their objective with intermediate service adjusted to keep total payment equal to the cap. The first branch remains at $1/4$. Hence it suffices to optimize $z \in [1/2, 3/4]$.

If $z > 1/2 + \delta$, Proposition EC.7.1 forces the middle branch to use the lower codeword deterministically. Its weighted loss alone is

$$\frac{3}{5} \{f(1/2) - f(1/4) + \frac{1}{2}(1/2 - 1/4)^2\} = \frac{21}{80}.$$

This exceeds the total feasible loss $3/160$ achieved by $(1/4, 1/2)$. Such a codebook is never optimal. For $z \in [1/2, \min\{3/4, 1/2 + \delta\}]$, the middle branch instead uses its mean-preserving adjacent lottery, while the highest branch takes deterministic terminal service z and intermediate service $3/4 - z$. Direct substitution

gives

$$\Psi(z) = \frac{z^2}{20} - \frac{z}{16} + \frac{3}{80}, \quad \Psi'(z) = \frac{z}{10} - \frac{1}{16}.$$

The unique unconstrained minimum is $z = 5/8$. Clamping it to the proven feasible interval gives $z_\delta = 1/2 + \min\{\delta, 1/8\}$. The one-code loss is $49/160$, larger than the two-code value throughout. The displayed formula is therefore the global at-most-two-code optimum. It reduces to $3/160$ at $\delta = 0$ and $23/1280$ for $\delta \geq 1/8$. $\square$

The last statement about being off-cap refers to the level itself, not a discrete number of feasible candidates. In particular $5/8$ remains off-cap for every larger δ as well. This example demonstrates a continuously moving operational design under a bounded realization policy. It is not a proof that the general bounded-overrun codebook problem has the same Monge structure as the expected-participation problem.

EC.8 Exact Nonsaturated Allocation and Supporting-Price Certificates

This section makes Proposition EC.4.4 computational: the targets, encoder, and a certificate are recovered with rational arithmetic for each fixed book. It also supplies a small-instance exact reference solver when allowable books come from a finite catalog. This does not discretize the unrestricted continuous problem or assert pathwise–deterministic equality away from saturation.

For quadratic primitives and fixed $c = (c_1, \dots, c_s)$, define adjacent slopes

$$\sigma_\ell = \frac{f(c_{\ell+1}) - f(c_\ell)}{c_{\ell+1} - c_\ell} = r - \frac{q}{2}(c_\ell + c_{\ell+1}), \quad 1 \leq \ell < s.$$

They are strictly decreasing and positive. The branch response in (EC.11) is affine with these slopes until c_s , and thereafter has derivative $-\gamma_j(t - c_s)$. Let $\mathcal{T}_j = [c_1, b_j]$ and assume $c_1 \leq \min_j b_j$ and $c_1 \leq B \leq \bar{b}$.

Proposition EC.8.1 (Exact allocation with a checkable supporting price). *A fixed-book optimizer and a rational multiplier λ can be obtained in $O(ks + k \log(k + 1))$ exact-rational arithmetic and comparison work. The result obeys*

$$\begin{aligned} \sum_j \pi_j t_j &= B, \quad t_j \in \mathcal{T}_j, \\ \sum_j \pi_j v_{j,c}(t_j) &= \lambda B + \sum_j \pi_j \max_{u \in \mathcal{T}_j} \{v_{j,c}(u) - \lambda u\}. \end{aligned} \tag{EC.14}$$

The right side is an upper bound for every feasible allocation, so equality certifies global fixed-book optimality. This remains valid with repeated caps, zero intermediate curvatures, singleton books, and endpoint promises. Recover terminal means $\min(t_j, c_s)$, adjacent mean-preserving tickets, and intermediate tiers $(t_j - c_s)_+$.

Proof. Initialize every target at c_1 . Increase the targets through successive chord intervals. At interval $[c_\ell, c_{\ell+1}]$, branch j offers capacity $\pi_j [\min(b_j, c_{\ell+1}) - c_\ell]_+$, all at the same marginal value σ_ℓ . Fill these capacities until the required aggregate promise is reached, using any fixed order to divide a tied marginal

segment. If the promise is reached in this interval, take $\lambda = \sigma_\ell$. Earlier intervals have larger marginal value, later ones have smaller marginal value, and capped coordinates need no further increase. Thus each chosen target maximizes its branch response minus λ times its target.

If all chord intervals are filled, $t_j = \min(b_j, c_s)$. Fill zero-curvature tails first at price zero. If additional promise is required, only positive-curvature tails remain. Their increments have the form

$$x_j = \min\{b_j - c_s, w/\gamma_j\}, \quad b_j > c_s, \gamma_j > 0,$$

with common water level $w \geq 0$ and $\lambda = -w$. Sort the cap events $w_j = \gamma_j(b_j - c_s)$. Between events the weighted total increment is affine in w , with positive slope $\sum_{j \text{ uncapped}} \pi_j/\gamma_j$. One rational division gives the required water level in the first interval that covers the remaining promise. Tied events are processed together or sequentially with zero intervening width. At the last event all tails are full, so the procedure covers $B = \bar{b}$ as well.

In every case the common multiplier supports every branch response on its feasible interval, proving (EC.14). The chord scan costs $O(ks)$, and the sorted tail sweep costs $O(k \log(k+1))$. Independently checking the certificate enumerates each branch's feasible codeword kinks, its two domain endpoints, and its clamped quadratic stationary point. There are $O(s)$ such candidates per branch, so checking does not change the bound. These operations preserve rationality. The bound counts exact-rational operations, not constant-time arithmetic on arbitrarily large integers. $\square$

Finite-catalog optimization at any promise. Enumerate feasible subsets of an input catalog of size N , with at most m levels. For each, apply Proposition EC.8.1 and subtract the charges of its actually selected levels. The largest certified value is the exact expected-participation catalog optimum at that promise. The work is at most

$$O\left(\sum_{s=1}^{\min(m,N)} \binom{N}{s} [ks + k \log(k+1)]\right).$$

This exponential reference method provides exact interior-promise comparisons; it is not the polynomial saturated recurrence of Theorem EC.9.1. A maximum over fixed-book concave value functions need not be concave. In particular, this procedure never swaps the outer codebook maximum with the inner supporting-price minimum. With nonnegative charges a redundant level can be deleted, but enumeration need not assume that all selected levels are used.

Independent validation. The implementation's primal chord and water-level sweep is compared with a separately coded dual-event search. That search scans all chord slopes, zero, and all negative tail-cap slopes, interpolating the aggregate target only between consecutive events. Both implementations are then checked by the finite candidate certificate in (EC.14). This checks the continuous branch maxima exactly, not only values on a target grid. At saturation, exhaustive charged-book allocation is also compared with the independently structured catalog path recurrence. Pairwise policy-transport tests exercise contractions and partial lifts under expected, pathwise, and deterministic timing, including intermediate tiers conditioned on drawn symbols. The original theorem proofs, not these finite tests, establish the all-input guarantees.

EC.8.1 What the shaped-charge example does and does not imply

Proposition B.3 establishes a logical separation for a specified nonmonotone charge. It does not establish that pathwise randomization typically saves engineering or certification cost. The following result gives a sharp institutional boundary for that example rather than assigning its specially shaped charge an unsupported operational interpretation.

Proposition EC.8.2 (One branch with a monotone charge). *For continuously chosen levels and a single branch, concave terminal reward, convex intermediate cost, and a nonnegative nondecreasing charge per selected level, a deterministic one-level policy weakly dominates every randomized policy at each feasible promise. This holds for every alphabet budget and every realization tolerance, including the full pathwise institution with draw-specific intermediate service.*

Proof. For an arbitrary feasible policy write $\mu = \mathbb{E}C$ and $\bar{y} = \mathbb{E}Y = B - \mu$. The deterministic level μ with intermediate service $\bar{y}$ attains total service $B \leq b$, and Jensen's inequalities improve both operating terms. The old book contains a largest used level $c_{\max} \geq \mu$. Nonnegativity and monotonicity imply that its total charge is at least $\rho(c_{\max}) \geq \rho(\mu)$. Thus the replacement is feasible, uses one level, and does not lower net payoff. $\square$

Consequently the one-branch separation in Proposition B.3 necessarily uses a charge outside this monotone class. The result covers zero charges, nondecreasing affine or convex charges, and a fixed opening fee plus a nondecreasing surcharge. It does not claim deterministic collapse for arbitrary heterogeneous multi-branch instances; the common alphabet couples their means. The original example and its proof remain intact as a mathematical possibility, not evidence of a generic operational advantage.

EC.9 Joint Design of a Risk-Limited Certified Alphabet

A finite certification catalog gives a general intermediate institution and a nontrivial codebook-dependent implementation problem. Let $\mathcal{A} = \{a_1 < \dots < a_N\} \subset [0, 1]$ be the allowable terminal levels, with $a_1 \leq b_1$. Selecting a level a incurs an arbitrary nonnegative charge $\rho(a)$; for example, it may combine a declared precision/storage charge and a per-level certification charge in the same units as service loss. These are stipulated prices, not estimated operational costs. A book is any subset of $\mathcal{A}$ of size at most m . The objective includes the charges of its actual selected levels, not merely its declared capacity.

The root promise is saturated in this section. Intermediate service y_j is chosen before the symbol draw, with

$$y_j + \mathbb{E}C_j = b_j, \quad y_j + C_j \leq b_j + \delta \quad \text{almost surely.} \quad (\text{EC.15})$$

This explicitly defined institution has the two familiar endpoints: $\delta = 0$ attains the pathwise/deterministic catalog optimum, while $\delta \geq 1$ removes the overrun restriction. The pre-draw restriction is not asserted without loss for arbitrary intermediate risk protocols.

For adjacent selected levels $u < v$, assign to this edge the branches $u \leq b_j < v$ and define

$$K_\delta(u, v) = \sum_{j: u \leq b_j < v} \pi_j \ell_j^\delta(u, v), \quad (\text{EC.16})$$

where, on that interval,

$$\ell_j^\delta(u, v) = \begin{cases} f(b_j) - \frac{v - b_j}{v - u} f(u) - \frac{b_j - u}{v - u} f(v), & v - b_j \leq \delta, \\ f(b_j) - f(u) + h_j(b_j - u), & v - b_j > \delta. \end{cases} \quad (\text{EC.17})$$

At a selected cap the branch loss is zero, including the boundary $b_j = u$. The last selected level u carries tail cost

$$T(u) = \sum_{j: b_j \geq u} \pi_j \{f(b_j) - f(u) + h_j(b_j - u)\}. \quad (\text{EC.18})$$

No original-cap anchoring is imposed on catalog entries.

Theorem EC.9.1 (Global catalog frontier with level-dependent charges). *For strictly increasing concave f and convex nondecreasing h_j , the following dynamic program globally minimizes service loss plus selected-level charges under the bounded-overflow institution:*

$$\begin{aligned} H_1(v) &= \begin{cases} \rho(v), & v \leq b_1, \\ +\infty, & v > b_1, \end{cases} \\ H_s(v) &= \rho(v) + \min_{u \in \mathcal{A}: u < v} \{H_{s-1}(u) + K_\delta(u, v)\}, \quad s \geq 2, \\ W_m^\delta &= \min_{1 \leq s \leq \min(m, N), v \in \mathcal{A}} \{H_s(v) + T(v)\}. \end{aligned} \quad (\text{EC.19})$$

Backtracking recovers the book, intermediate tiers, and lottery probabilities. All budgets through m require $O(kN^2 + mN^2)$ arithmetic/evaluation work with a direct branch-sum precomputation, and $O(N^2 + mN)$ storage. For rational quadratic data and rational catalog charges the calculation is exact in rational arithmetic. An arbitrary additional charge depending only on s can be added at the final minimization, but level charges must enter the recurrence itself.

Proof. For a fixed book, Proposition EC.7.1 gives the optimal branch rule. On $u \leq b_j < v$, if $v - b_j \leq \delta$ use the adjacent unbiased lottery of mean b_j and set $y_j = 0$; otherwise use u deterministically and set $y_j = b_j - u$. Branches above the last level also use that last level deterministically. Consequently the book's exact loss is the sum of its adjacent-edge costs and its terminal tail. Every branch is counted exactly once; cap endpoints contribute zero regardless of the interval convention.

Increasing paths through the catalog enumerate every possible sorted book. A path can begin exactly at levels no larger than b_1 , which is necessary and sufficient for feasibility. Charging ρ at each visited vertex adds the book's actual implementation cost. Induction on path length gives (EC.19); appending the tail and minimizing over feasible lengths proves both inequalities needed for global optimality. If a selected level is unused, deleting it weakly lowers its nonnegative charge and does not worsen service. Thus the at-most-

budget optimization also covers redundant symbols. Store predecessor indices and replay the fixed-book rule to reconstruct a controller.

There are $O(N^2)$ edges, each computable by k branch evaluations, and m layers of $O(N^2)$ comparisons/additions. Storing edges and backpointers gives the stated bounds. In the quadratic rational model every interpolation, difference, and cost is rational. There is no continuous optimizer or rounding tolerance in the recurrence. At $\delta = 0$ every strictly bracketing upper level is excluded, so the branch rule is the deterministic one; at $\delta \geq 1$ every possible upper level is admissible. The endpoint claims follow after optimization over the same catalog. $\square$

This theorem solves the saturated catalog specialization of the unified design problem for arbitrary k, N, m, δ and nonuniform ρ . An exact catalog optimum is not itself an optimum over all real codewords; Theorem E.1 supplies the missing quantitative comparison. This recurrence does not assume that risk-limited costs are Monge. The continuous uncharged expected frontier retains its separate $O(mk)$ guarantee; the certified-catalog theorem uses a general ordered-path algorithm to accommodate actual level charges and risk eligibility. Entropy coding, shared-table compression depending on pairs of levels, repeated-customer state, and arbitrary codebook-dependent costs are not silently reduced to additive ρ .

Why a loss frontier followed by a resource surcharge is insufficient. Use the off-cap model of Proposition C.3 and catalog $\{1/4, 1/2, 5/8, 3/4\}$ with two symbols and $\delta = 1$. With zero charges, $(1/4, 5/8)$ attains loss $23/1280$. Give only level $5/8$ charge $1/640$. That book now costs $25/1280$, whereas each cap-only pair $(1/4, 1/2)$ and $(1/4, 3/4)$ costs $24/1280$. The optimum changes its selected levels, not merely its capacity. The exact catalog recurrence finds the change; retaining the uncharged loss-minimizing book and adding its storage bill does not.

Risk exposure along the continuous example. The original two-symbol continuous example is retained as an exact illustration beyond a fixed catalog. Its globally optimal top level and loss are

$$z_\delta = 1/2 + \min(\delta, 1/8), \quad D_2^\delta = z_\delta^2/20 - z_\delta/16 + 3/80. \quad (\text{EC.20})$$

On the middle branch the high-realization probability is

$$p_{\text{high}}(\delta) = \frac{1/4}{z_\delta - 1/4}, \quad \text{maximum overrun} = z_\delta - 1/2.$$

At zero tolerance this branch uses its cap deterministically; at tolerance $1/8$, the high probability is $2/3$ and the possible overrun is $1/8$. Thus probability, mean obligation, and maximum overrun are separate reported quantities. Figure 1 shows the continuous loss profile; the complete global-example proof is in the electronic companion.

EC.10 From Price Bounds to a Fixed Expanded Alphabet

The polynomial price oracle also gives a constructive guarantee when the execution interface can accommodate additional levels. This result does not require the outer price gap to vanish, nor does it randomize the installed alphabet. A single union alphabet is installed before execution, and its branch rules retain the original promise and realization constraints.

Use the ordered-interface class of Section A. Let $R(c) = \sum_{z \in c} \rho(z)$, $\bar{b} = \sum_j \pi_j b_j$, and $H = L_f + \max_j H_j + 2$, where L_f and H_j are Lipschitz bounds from Section E. A priced optimizer consists of a book c and targets t_j ; write $S = \sum_j \pi_j t_j$ and $J = \sum_j \pi_j v_{j,c}(t_j) - R(c)$. Thus its exact price value is $D(\lambda) = J + \lambda(B - S)$. The feasible-first-level restriction in (44) is retained even when $S \neq B$.

Theorem EC.10.1 (Two-price recovery with resource augmentation). *Suppose the catalog is feasible, $\rho \geq 0$, and $\varepsilon > 0$. With an exact scalar shortfall oracle, at most*

$$2 + \max \left\{ 0, \left\lceil \log_2 \frac{H(\bar{b} - a_1)}{4\varepsilon} \right\rceil \right\} \quad (\text{EC.21})$$

price calls suffice, interpreting the maximum as zero when $\bar{b} = a_1$. The algorithm returns an original-budget upper bound U , and one fixed alphabet c^U of size at most $\min(2m, N)$ with exactly feasible targets and net payoff J_U such that

$$J_U \geq U - e - \Omega \geq \mathcal{V}_{m,\mathcal{A}}^\delta(B; \rho) - \varepsilon - \Omega, \quad 0 \leq e \leq \varepsilon. \quad (\text{EC.22})$$

Here the renewal notation denotes its specialization; the same result holds for the abstract class. The reported charge excess Ω is computed from the two selected books, not replaced by an unobserved constant. With zero charges $\Omega = 0$. For charges bounded by $\rho_{\max}$, $0 \leq \Omega \leq m\rho_{\max}$.

For rational quadratic data the algorithm is exact rational arithmetic and uses $O((k + m)N^2[1 + \log_+(H(\bar{b} - a_1)/\varepsilon)] + k \log(k + 1))$ arithmetic operations, where $\log_+ x = \max(0, \log_2 x)$ with value zero at $x = 0$. Its storage is $O(N^2 + mN + km)$, and its bit complexity is polynomial in the encoded input and the number of price calls. If $|c^U| \leq m$, (EC.22) is also a same-budget certificate; otherwise J_U must not be used as a lower endpoint for the original-budget optimum.

Construction and proof outline. Start with prices $\lambda_- = -\max_j H_j - 1$ and $\lambda_+ = L_f + 1$. Their optimizing totals satisfy $S_- \geq B \geq S_+$. Bisect prices, retaining this bracket, until its width times $(\bar{b} - a_1)/4$ is at most ε . An exact optimizer with total B already gives an original-budget global certificate. Otherwise $S_- > B > S_+$, and set

$$\theta = \frac{B - S_+}{S_- - S_+}, \quad c^U = c^- \cup c^+, \quad t_j^U = \theta t_j^- + (1 - \theta)t_j^+, \quad (\text{EC.23})$$

$$e = (\lambda_+ - \lambda_-) \frac{(S_- - B)(B - S_+)}{S_- - S_+}, \quad \Omega = R(c^U) - \theta R(c^-) - (1 - \theta)R(c^+). \quad (\text{EC.24})$$

Install c^U once and apply its canonical response to t_j^U . Enlarging a book weakly improves its response on the original target domain. Concavity of that response then bounds the new operating payoff below by the

convex combination of the two operating payoffs. This is a feasible-policy construction, not an assumption that the original book problem is convex. Also,

$$\theta D(\lambda_-) + (1 - \theta) D(\lambda_+) = \theta J_- + (1 - \theta) J_+ + e.$$

Take $U = \min\{D(\lambda_-), D(\lambda_+)\}$ and subtract the actual union charge. The elementary bound $e \leq (\lambda_+ - \lambda_-)(\bar{b} - a_1)/4$ proves termination and accuracy. The full endpoint, charge, risk, and bit-complexity arguments appear in Section EC.11 of the companion. $\square$

Lagrangian convexification and recovery from supporting solutions are classical (Handler and Zang, 1980; Lemaréchal, 2001). The structural addition is that, in this ordered-interface problem, their aggregate-target mixture can be implemented by a *fixed* union alphabet and one pre-draw intermediate tier on each branch. Concavity after eligibility truncation is essential: averaging terminal levels can break catalog membership, while mixing services and lotteries directly need not respect the pre-draw service restriction. Equations (EC.23)–(EC.24) avoid both problems and charge every installed level.

Corollary EC.10.2 (Continuous comparison with variable alphabet budget). *For a catalog of maximum spacing h , containing zero and one, the same algorithm returns an exactly feasible, at-most- $2m$ -level policy satisfying*

$$J_U \geq \mathcal{V}_m^\delta(B; \rho) - K_m h - \varepsilon - \Omega.$$

In particular, with zero charges, a uniform mesh with denominator $\max\{1, \lceil 2K_0/\eta \rceil\}$ and price accuracy $\varepsilon = \eta/2$ gives payoff at least $\mathcal{V}_m^\delta(B; 0) - \eta$. For rational quadratic primitives this is polynomial in k , m , K_0/η , and the sampled rational encoding length. It preserves the promise and risk ceiling without slack.

Proof. Combine Theorems E.1 and EC.10.1; $K_0 = L_f + \sum_j \pi_j H_j$ is independent of the alphabet budget. $\square$

This is an additive resource-augmentation guarantee, not an FPTAS at the original memory limit. The exact original-budget continuous theorem and the finite prefix search remain available when additional symbols cannot be installed. With charged levels, the explicit Ω identifies the economic price of the augmentation instead of concealing it inside an operating-payoff comparison.

EC.11 Detailed Recovery Proof and a Strict Price Gap

Endpoint brackets. For every feasible book, the branch response on $[c_1, b_j]$ has slopes between $-H_j$ and L_f : the increasing interpolant has slopes in $[0, L_f]$, while the singleton tail has slopes in $[-H_j, 0]$. Hence at $\lambda_- = -\max_j H_j - 1$ the priced response is strictly increasing, and every branch chooses b_j . At $\lambda_+ = L_f + 1$ it is strictly decreasing, and every branch chooses $c_1 \leq B$. This proves the initial total-target bracket for any optimizing book. It applies at $B = 0$, at $B = \bar{b}$, and at a singleton target domain. If $\bar{b} = a_1$, both endpoints already have total B .

At every midpoint, use an exact priced optimizer and replace the left endpoint when its total exceeds B , or the right endpoint when its total is below B . At equality, its exact promise makes its priced objective equal its

feasible objective; weak duality certifies original-budget global optimality. Ties between optimizing books or within a branch do not obstruct the rule: every selected optimizer gives the stated bound and belongs to one of the three total-target cases. After q bisections the retained width is $H2^{-q}$. Thus the call bound in (EC.21) holds without solving an algebraic equation for an exact dual minimizer.

Union reconstruction. Let $c^U = c^- \cup c^+$, with its levels sorted once. At any target feasible for c^- , its old branch rule is feasible under c^U with unchanged intermediate tier and support. The union response, being optimal for that target, weakly dominates it. The same holds for c^+ . Every mixed target lies in the union domain $[\min c^U, b_j]$. Concavity of the union response gives

$$v_{j,c^U}(t_j^U) \geq \theta v_{j,c^U}(t_j^-) + (1 - \theta) v_{j,c^U}(t_j^+) \geq \theta v_{j,c^-}(t_j^-) + (1 - \theta) v_{j,c^+}(t_j^+). \quad (\text{EC.25})$$

For a branch with highest union-eligible level d_j^U , choose the fixed intermediate tier $(t_j^U - d_j^U)_+$ and the adjacent union lottery of mean $\min(t_j^U, d_j^U)$. If the tier is positive, the lottery is a singleton and total realized service equals $t_j^U \leq b_j$. Otherwise the tier is zero and every support point is at most its eligibility threshold. Thus the renewal realization constraint holds exactly, including $\delta = 0$. The tier is fixed *before* the draw; this construction does not smuggle in a random tier, a random installed book, or an uncharged shared random seed. The same argument proves the abstract ordered-interface statement using its target and support constraints. The weighted sum of mixed targets is exactly B .

Payoff and charge accounting. Write $J_{\text{mix}} = \theta J_- + (1 - \theta) J_+$. Since $\theta(S_- - B) = (1 - \theta)(B - S_+)$, expanding the weighted price values proves the identity in the main proof. Their minimum U is at most that weighted value and remains an upper bound for every original-budget feasible book. Therefore $J_{\text{mix}} \geq U - e$. Put $x = S_- - B$ and $y = B - S_+$; both are positive, and $xy/(x + y) \leq (x + y)/4 \leq (\bar{b} - a_1)/4$. This proves the reported error bound, using the actual targets when available rather than only their total range.

Let R_- , R_+ , and $R_\cap$ denote the sums of charges in the two books and their intersection. Inclusion-exclusion gives

$$\Omega = (1 - \theta)R_- + \theta R_+ - R_\cap = (1 - \theta)R(c^- \setminus c^+) + \theta R(c^+ \setminus c^-) \geq 0.$$

It is at most $m\rho_{\max}$ and is zero for uncharged levels. Subtracting $R(c^U)$ from (EC.25), after weighting and summing, proves $J_U \geq J_{\text{mix}} - \Omega \geq U - e - \Omega$. If an endpoint has total exactly B , use it twice, set $\theta = 1$, and put $e = \Omega = 0$.

The union may have fewer than $2m$ levels because of overlap. If it has at most m , it is feasible for the original budget and its value can be included in the original lower bound. Otherwise the interval for that budget must use a separately feasible m -level policy. In our implementation this policy is obtained by reallocating the exact promise on each endpoint book with the fixed-book allocator and retaining the better value. That lower bound does not rely on a vanishing price gap.

Arithmetic and storage. Each price call uses Theorem A.1. For quadratic costs, a branch-support maximizer is found at an eligible knot, the cap, or a clipped tail stationary point, in $O(km)$ rational work after the book has been recovered. Recovering two endpoint books and the union rules also takes $O(km)$ work. The original-budget feasible lower policies use $O(km + k \log(k + 1))$ additional work each; the additional $k \log(k + 1)$ term in the main theorem accounts for sorting its heterogeneous tail events. This optional lower-bound calculation can be omitted when only the augmentation guarantee is needed. Retaining only the two endpoint policies, rather than all intermediate tables, gives $O(N^2 + mN + km)$ storage. Bisection adds at most one bit per call to a common denominator. Scalar quadratic evaluations, sums over polynomially many rational terms, interpolation, and the final ratio defining θ have bit lengths polynomial in the input and the call count. The continuous-mesh corollary states dependence on the numerical scale K_0/η explicitly; it does not turn a magnitude-dependent mesh into a strongly polynomial algorithm.

EC.11.1 A fixed instance with a strictly positive outer price gap

Take two equally weighted branches, caps $(1/4, 3/4)$, quadratic coefficients $r = 2, q = 1, \gamma = (1, 2)$, promise $B = 2/5$, risk tolerance zero, budget two, and the catalog $\{i/8 : 0 \leq i \leq 8\}$ with zero charges. The exact original-budget optimum is $45/64$. Its book $(1/4, 5/8)$ uses targets $(1/4, 11/20)$; the second branch assigns probabilities $1/5$ and $4/5$ to its two levels.

For completeness, any singleton book or any book whose highest used eligible level is at most $1/4$ has operating value at most $f(1/4) = 15/32 < 45/64$. An upper codeword above $3/4$ is useless under pathwise participation and can be deleted. The remaining two-level possibilities have lower level $0, 1/8$, or $1/4$, and upper level $3/8, 1/2, 5/8$, or $3/4$. Direct substitution in the concave branch allocation gives the following exhaustive values:

Lower level \ upper level	3/8	1/2	5/8	3/4
0	1977/6400	169/400	10027/19200	487/800
1/8	2827/6400	10627/19200	6221/9600	43/64
1/4	3577/6400	1073/1600	45/64	111/160

For an entry, eliminate t_2 as $4/5 - t_1$, evaluate the two piecewise-quadratic responses, and maximize on the feasible interval. This yields the displayed rational values and proves the global claim without trusting a search-order heuristic.

At price $23/16$, the optimal priced books $(1/4, 5/8)$ and $(1/4, 1/2)$ have totals $7/16$ and $3/8$, with net values $195/256$ and $43/64$. At this positive price, intermediate service has nonpositive priced payoff. The maximum eligible values of $f(c) - (23/16)c$ are $7/64$ on branch one and $5/32$ on branch two. Consequently every book has price bound at most $(23/16)(2/5) + (7/64 + 5/32)/2 = 453/640$, and both displayed books attain it. Thus $D(23/16) = 453/640$ independently of the recurrence implementation. Their mixture with weight $2/5$ on the first book has total $2/5$ and value $453/640$. For any price, the maximum priced value is at least the corresponding weighted value, which equals this mixture value because its promise is exact. Hence

$$\inf_{\lambda} D(\lambda) = \frac{453}{640}, \quad \inf_{\lambda} D(\lambda) - \mathcal{V}_{2,\mathcal{A}}^0(2/5; 0) = \frac{3}{640} > 0.$$

The fixed union alphabet $(1/4, 1/2, 5/8)$ attains $453/640$: branch one uses $1/4$; branch two draws $1/2$ with probability $3/5$ and $5/8$ with probability $2/5$. No intermediate service is needed, and both risk bounds hold. The example demonstrates why increased price accuracy alone cannot force the original-budget gap to zero, while the union construction removes this particular obstruction using one extra level. It is not a lower bound proving that the factor two is necessary for every instance.

EC.12 Joint Approximation Without Additional Memory

The exact conditional oracle permits a global approximation of the original catalog problem without enumerating books or increasing their cardinality. The price is dependence on the dimension of the target vector. Let f be L_f -Lipschitz and each h_j be $L_{h,j}$ -Lipschitz on $[0, b_j]$, and set $L = \max\{L_f, \max_j L_{h,j}\}$. For the quadratic implementation, $L = \max\{r, \max_j \gamma_j\}$ is valid. These assumptions bound value changes, not the sensitivity of a selected alphabet.

Lemma EC.12.1 (Target sensitivity and exact repair). *For every feasible book c , its branch response is L -Lipschitz on $[c_1, b_j]$. Given $u_j \in [c_1, b_j]$ and a feasible root promise B , bounded transfers produce $t_j \in [c_1, b_j]$ with $\sum_j \pi_j t_j = B$ and*

$$\sum_j \pi_j |t_j - u_j| = \left| B - \sum_j \pi_j u_j \right|. \quad (\text{EC.26})$$

The operating payoff decreases by at most L times the right-hand side. With rational inputs, both transfers and canonical lotteries are rational and satisfy the original constraints exactly.

Proof. Chord slopes of f have absolute value at most L_f ; the final shortfall tail has slopes bounded by $L_{h,j}$. Their continuous concatenation is therefore L -Lipschitz. If the residual is positive, increase targets within their remaining cap capacities; if it is negative, decrease them toward c_1 . Total available capacity is sufficient because $B \in [c_1, \bar{b}]$. Stop when the residual is zero. All changes have the same sign, proving (EC.26); the payoff bound follows by the weighted Lipschitz inequality. Reconstruct the lotteries after repairing targets, rather than rounding their probabilities separately. The canonical rule then enforces $y_j + C_j \leq \tau_j$ for each realization. $\square$

Theorem EC.12.2 (Same-budget additive scheme at fixed branch count). *For a feasible finite catalog and any $\varepsilon > 0$, a target-net algorithm returns one feasible book of at most m levels and net value J satisfying*

$$\mathcal{V}_{m,\mathcal{A}}^\tau(B; \rho) - \varepsilon \leq J \leq \mathcal{V}_{m,\mathcal{A}}^\tau(B; \rho). \quad (\text{EC.27})$$

For $k \geq 2$, set $\eta = \varepsilon / [2L(k-1)]$. At most

$$N(1 + 1/\eta)^{k-1} \quad (\text{EC.28})$$

target profiles need be optimized. Thus a safe arithmetic bound is $O((k+m)N^3(1+2L(k-1)/\varepsilon)^{k-1})$. For $k = 1$, apply the conditional oracle once at $t_1 = B$. The guarantee allows arbitrary nonnegative catalog

charges and arbitrary ceilings $\tau_j \geq b_j$; neither is approximated. For rational quadratic data the algorithm is polynomial in input bit length and the displayed numerical accuracy parameter when k is fixed. It is not an FPTAS with variable k and does not claim polynomial dependence on the binary length of $1/\varepsilon$.

Proof. Enumerate a possible lowest level $a \in \mathcal{A}$ with $a \leq a_* := \min\{B, \min_j b_j\}$. Restrict the conditional candidate pool to levels at least a . In weighted target coordinates, let $x_j = \pi_j(t_j - a)$, $d_j = \pi_j(b_j - a)$, and $C = B - a$. Choose one distinguished branch, numbered k only for this proof. Enumerate multiples of η in $[0, d_j]$ for $j < k$, discarding tuples whose sum exceeds C . Set

$$x_k = \min\{d_k, C - \sum_{j < k} x_j\}.$$

Any remaining deficit is nonnegative; fill it by bounded increases to the coordinates $j < k$. Their total available capacity is sufficient because $C \leq \sum_j d_j$. This constructs a feasible profile $t_j = a + x_j/\pi_j$. Run Theorem 3.1 at budget m and retain the best policy over all anchors and profiles.

Let (c^*, t^*) be a global catalog optimum, which exists by compactness of each fixed-book allocation problem and finiteness of the catalog. Its lowest level $a = c_1^*$ is enumerated. For $j < k$, the enumeration includes $x_j = \eta \lfloor x_j^*/\eta \rfloor$. Their aggregate decrease is $D < (k-1)\eta$. The constructed last coordinate is at least x_k^* and increases by at most D . If it reaches its cap, the subsequent increases on the other coordinates use only the remaining deficit. Consequently, the final weighted target distance from t^* is at most $2D \leq 2(k-1)\eta$. All final targets remain at least a , so the *same* book c^* is feasible. Lemma EC.12.1 bounds its payoff loss by ε ; its opening charge is unchanged. The conditional optimizer does no worse. Maximizing over profiles proves (EC.27).

Each of the $k-1$ coordinates has at most $1 + d_j/\eta \leq 1 + 1/\eta$ possibilities. There are at most N anchors. Applying the conditional-oracle complexity gives the stated bound. The construction uses rational flooring, addition, comparison, and the rational oracle, so the bit-complexity conclusion follows under the stated representation assumptions. $\square$

The scheme is useful when the number of distinct contract branches is small or repeated homogeneous branches have been aggregated. Its exponential accuracy dependence on k is not concealed by the polynomial dependence on the symbol budget. At a large branch count we use selected target profiles and certified upper bounds rather than declare the net completed. An interrupted enumeration returns a feasible incumbent, but the implementation deliberately withholds the requested ε guarantee until coverage is complete.

This result addresses the original-budget difficulty differently from union recovery. It pays for accuracy in target-space computation, not extra installed levels. At saturation, Corollary 3.2 avoids target-space enumeration altogether. For a general interior promise, selected profiles can be polished by alternating exact inner allocation and exact conditional book design. This preserves feasibility and nondecreases value at every step, but a fixed iteration cap or stationary pair has no global-optimality implication. The comparative study records this distinction explicitly.

EC.13 Charge-Aware Recovery from Inexact Price Supports

The polynomial price oracle in Section A remains a useful source of target profiles and upper bounds. We now state exactly what survives an inexact oracle, and how a fixed memory budget enters the recovery guarantee. For a feasible endpoint policy $i \in \{-, +\}$, let c^i , t^i , J_i , and $S_i = \sum_j \pi_j t_j^i$ be its book, targets, net payoff, and aggregate target. It need not satisfy the desired root promise B , but satisfies all branch constraints. Its book has at most m levels and its lowest level is at most a_* . At prices $\lambda_- \leq \lambda_+$ assume certified bounds $u_i \geq D(\lambda_i)$, and write

$$\zeta_i = u_i - \{J_i + \lambda_i(B - S_i)\} \geq 0. \quad (\text{EC.29})$$

This defect includes both optimization error and conservatism in the certified upper bound. An approximate feasible solution alone, without an upper bound, does not certify ζ_i .

Suppose $S_- > B > S_+$. Define

$$\theta = \frac{B - S_+}{S_- - S_+}, \quad t^U = \theta t^- + (1 - \theta)t^+, \quad c^U = c^- \cup c^+, \quad R_{\text{mix}} = \theta R(c^-) + (1 - \theta)R(c^+),$$

and

$$e = (\lambda_+ - \lambda_-) \frac{(S_- - B)(B - S_+)}{S_- - S_+}, \quad U = \min\{u_-, u_+\}. \quad (\text{EC.30})$$

Here U is the minimum of two valid endpoint bounds, not necessarily the minimized Lagrangian dual value. Endpoint equality can be handled by returning that endpoint policy; the strict bracket avoids division by zero.

Theorem EC.13.1 (Budgeted compression with oracle defects). *For any feasible sub-book $d \subseteq c^U$ with $|d| \leq s$ at the fixed mixed targets, put*

$$\Delta(d) = \sum_j \pi_j \{v_{j,c^U}(t_j^U) - v_{j,d}(t_j^U)\} \geq 0, \quad \Gamma_s = \min_d \{\Delta(d) + R(d) - R_{\text{mix}}\}.$$

Theorem 3.1 computes the minimizer and its policy. Its net payoff satisfies

$$J_s \geq U - e - \theta \zeta_- - (1 - \theta) \zeta_+ - \Gamma_s. \quad (\text{EC.31})$$

The policy has at most s symbols and preserves the mixed branch targets, their aggregate root promise, and every realization ceiling. The original budget is obtained by setting $s = m$. Optimizing over the full catalog instead of the union, or returning a better feasible incumbent at the same budget, preserves the bound. For rounded mixed targets, let r_{mix} bound their weighted distance from t^U after clipping to the selected book extquotesingles feasible target intervals, and let r_{eq} be their root residual. A further allowance $L(r_{\text{mix}} + r_{\text{eq}})$ suffices after exact repair.

Proof. The union weakly improves the branch response at each endpoint target because it contains both old books. Its response is concave in the target, so

$$\sum_j \pi_j v_{j,c^U}(t_j^U) \geq \theta [J_- + R(c^-)] + (1 - \theta) [J_+ + R(c^+)].$$

This compares operating values before subtracting installation charges. Direct expansion of (EC.29) gives

$$\theta u_- + (1 - \theta)u_+ = \theta J_- + (1 - \theta)J_+ + e + \theta \zeta_- + (1 - \theta)\zeta_+.$$

The left side is at least U . Subtracting $\Delta(d) + R(d)$ from the union operating value proves (EC.31) for any d , and minimizing that subtraction is precisely fixed-target book optimization. A feasible singleton always exists in the union: its smallest level is no higher than either endpoint's targets, hence no higher than each mixed target. Canonical reconstruction proves all feasibility claims. The repair loss follows by first bounding the target perturbation and then applying the previous lemma to the residual; installed-level charges do not change. $\square$

Unlike the union excess charge, Γ_s includes an *optimized* distortion–charge tradeoff at the actual returned budget. It may be positive, zero, or negative, and the bound can still be loose if compression is costly. We do not relabel it as a uniform relative approximation. The independently available net scheme gives the unconditional additive same-budget guarantee at fixed branch count; the compressed-support bound provides a computable instance-specific certificate at large branch counts.

For exact price optimizers, $\zeta_i = 0$ when their price bounds are exact. Optimality at two prices gives the explicit tie-safe monotonicity argument

$$J_1 - \lambda_1 S_1 \geq J_2 - \lambda_1 S_2, \quad J_2 - \lambda_2 S_2 \geq J_1 - \lambda_2 S_1 \quad \implies \quad (\lambda_2 - \lambda_1)(S_1 - S_2) \geq 0.$$

Arbitrary selections among exact tied optima therefore preserve the ordering of supported totals. Approximate optimizers need not do so. The inexact theorem assumes and checks a feasible bracket rather than extending this monotonicity assertion incorrectly. It establishes a value bound conditional on certified endpoint data, not an oracle-call complexity bound for an unspecified approximate solver.

EC.13.1 Relation to union augmentation and its strict gap

At $s = |c^U|$, choosing the whole union gives $\Delta = 0$ and $R(c^U) - R_{\text{mix}} = \Omega$; (EC.31) recovers the charged union guarantee. For zero charges and exact supports, the usual price-bracket tolerance produces an at-most- $2m$ policy within the requested additive tolerance of the original-budget optimum. The full construction and continuous mesh transfer remain in the companion. They are bicriteria results, distinct from the original-budget net scheme and saturation theorem.

The exact two-branch example has caps $(1/4, 3/4)$, equal weights, curvatures $(1, 2)$, $f(x) = 2x - x^2/2$, catalog $\{i/8 : 0 \leq i \leq 8\}$, $m = 2$, $B = 2/5$, and $\delta = 0$. Its original-budget value is $45/64$, whereas the minimum price bound is $453/640$. The gap $3/640$ is $1/150$ of the original value. A three-level book $(1/4, 1/2, 5/8)$ attains the price value. This is a relaxation-gap example, not evidence that the factor two is necessary. The conditional frontier quantifies every compression budget for its mixed targets; the joint net scheme can instead approximate the two-level optimum. No factor-two necessity claim or same-budget strong-duality claim is used anywhere in these arguments.

EC.14 Local Distortion and Same-Budget Certificates

The target net provides a global guarantee when fully enumerated. A complementary certificate can be evaluated with just one conditional dynamic program. Let

$$t_j^0 = \min\{b_j, z\}, \quad \sum_j \pi_j t_j^0 = B, \quad W_0 = \sum_j \pi_j f(t_j^0).$$

At $B = \bar{b}$ take $t^0 = b$. The water level is otherwise found by sorting caps. Let $\rho_* = \min_{a_i \leq a_*} \rho_i$, where $a_* = \min\{B, \min_j b_j\}$. Feasibility guarantees that this set is nonempty.

Theorem EC.14.1 (Distortion–charge certificate). *For every symbol budget $s \geq 1$, define*

$$C_s = \min_{\substack{c \subseteq \mathcal{A}, |c| \leq s \\ c_1 \leq \min_j t_j^0}} \left[\sum_j \pi_j \{f(t_j^0) - v_{j,c}(t_j^0)\} + R(c) \right].$$

The conditional dynamic program computes $C_s = W_0 - P_s(t^0)$ exactly, and

$$P_s(t^0) \leq \mathcal{V}_{s,\mathcal{A}}^\tau(B; \rho) \leq W_0 - \rho_*. \quad (\text{EC.32})$$

Thus $C_s - \rho_$ is an explicit original-budget optimality-gap certificate. Any other valid upper bound for that budget can replace the right-hand side by its minimum with $W_0 - \rho_*$. A larger-budget policy value is not such an upper bound.*

Proof. For any policy, $\mathbb{E}C_j \leq t_j$ and Jensen’s inequality and monotonicity give $\mathbb{E}f(C_j) - h_j(y_j) \leq f(t_j)$. Maximizing the sum of these ideal rewards under the root equality and caps gives t^0 : their marginal rewards are equal on uncapped branches, and weakly larger on capped branches. This remains an optimum when f is not strictly concave. Every feasible nonempty book contains a level at most a_* and pays at least ρ_* . Hence its net value is at most $W_0 - \rho_*$. Conversely, the conditional optimal policy at t^0 is feasible at budget s . Finally, minimizing the displayed distortion plus charge is exactly the conditional objective subtracted from W_0 . $\square$

The certificate remains valid with nonuniform or large fees, because those fees are optimized rather than charged after an uncontrolled expansion. It can still be loose: C_s includes unavoidable finite-memory distortion, the cost of the selected levels, and any loss caused by fixing t^0 . We report its numerical size instead of asserting that it is uniformly small relative to the optimum. A multiplicative guarantee is particularly inappropriate when the optimal net payoff is zero or negative.

EC.14.1 Local errors, adaptive candidates, and constructive rates

For a twice differentiable f , suppose $-f'' \leq M_{uv}$ on $[u, v]$. If an eligible pair brackets a target t , its interpolation error satisfies

$$0 \leq f(t) - \left[f(u) + \frac{t-u}{v-u} (f(v) - f(u)) \right] \leq \frac{M_{uv}}{2} (t-u)(v-t). \quad (\text{EC.33})$$

This follows by comparing f with its chord plus $M_{uv}(t - u)(v - t)/2$, whose difference has nonnegative second derivative and zero endpoint values. If the next selected level is ineligible or no upper level exists, the exact local deficit is

$$f(t) - f(u) + h_j(t - u), \quad (\text{EC.34})$$

not a second-order chord error. Summing these local quantities with the actual selected-level charges yields C_s for the chosen book. A piecewise curvature majorant can also be placed directly in the target-crossing recurrence to minimize a certified distortion bound. It must retain the first-order expression (EC.34) on ineligible edges.

An adaptive candidate pool can include ideal target values and refine intervals that matter for (EC.33)–(EC.34). The resulting certificate is evaluated from the final pool and policy. We make no uniform convergence-rate claim for the greedy midpoint refinement used in the experiments. In particular, adding a level just above an eligibility threshold need not help the affected branch. Candidate selection and the exact optimization over those candidates are distinct tasks.

Proposition EC.14.2 (Original-budget comparison books). *Let $m \geq 2$, $h = 1/(m - 1)$, and suppose the catalog contains the uniform m -point book $G = \{0, h, \dots, 1\}$. With zero charges, the conditional policy at t^0 satisfies*

$$\mathcal{V}_{m,\mathcal{A}}^\tau(B; 0) - P_m(t^0) \leq L_f h + \sum_j \pi_j h_j(h), \quad (\text{EC.35})$$

$$\mathcal{V}_{m,\mathcal{A}}^\tau(B; 0) - P_m(t^0) \leq M h^2/8 \quad \text{if } \min_j(\tau_j - b_j) \geq h \text{ and } -f'' \leq M. \quad (\text{EC.36})$$

In (EC.35), $h_j(h)$ may be replaced by $h_j(\min\{h, b_j\})$ if costs are defined only on $[0, b_j]$. The same bounds hold relative to the uncharged continuous m -level optimum. With charges, adding $R(G) - \rho_*$ to the right-hand side gives the corresponding catalog comparison. If the distinct ideal targets are themselves available and number at most m , the uncharged policy attains W_0 and is globally optimal, for any realization headroom.

Proof. At each target, round its terminal level down to the nearest point of G and supply the difference before execution. The difference is at most h , the target and root promise are unchanged, and every realized total equals the target, hence is no larger than b_j . This proves (EC.35) by comparison with W_0 . Under the headroom assumption the upper neighbor is no higher than $t_j^0 + h \leq b_j + h \leq \tau_j$, so the adjacent unbiased lottery is feasible with no intermediate service. Apply (EC.33) and $(t - u)(v - t) \leq h^2/4$. The conditional optimum on the whole catalog is no worse than either comparison policy. Since W_0 also bounds continuous design, the continuous comparison follows. With fees, the comparison book pays $R(G)$ while the catalog upper bound subtracts ρ_* . Finally, installing the distinct ideal targets realizes them deterministically with no intermediate service, attaining the ideal upper bound. $\square$

These are explicit same-budget guarantees, not claims that uniform catalogs are always efficient. Their approximation parameter is the number of installed symbols, whereas Theorem EC.12.2 controls computational accuracy at a fixed symbol budget. The local certificate and the direct continuous comparisons

below distinguish the two effects. In particular, driving a price tolerance to zero cannot remove finite-catalog distortion or the original-budget price gap.

EC.15 Pricing Writable Capacity and Other Implementation Resources

Writable state, read-only constants, evaluation work, and compilation are different resources. A finite menu of implementations t for institution I , with charges depending only on capacity m , can be compared by

$$\min_{I,m,t} \{D_m^I + \lambda_w \lceil \log_2 m \rceil + \lambda_R R_{I,t}(m) + \lambda_T Q_{I,t}(m) + C_{I,t}(m)/H_0 + \kappa_I\}. \quad (\text{EC.37})$$

Here H_0 is the number of repeated deployments over which compilation is amortized, and κ_I is an exogenous protocol charge. This is accounting followed by finite comparison, not an additional structural optimization theorem. Its validity requires the stipulated menu charges to be independent of the particular book at a given capacity. Theorem EC.9.1, in contrast, puts actual additive level charges inside book selection.

With only a writable-bit price, capacity 2^w uses the frontier at $\min(2^w, k)$. For the equal-probability three-cap example, the zero-bit loss is $7/16$, the one-bit losses are $1/96$ under expected and $1/8$ under pathwise participation, and two bits attain full information. Thus a bit price strictly between $1/96$ and $1/8$ selects one bit under expected participation and two under pathwise participation when all other protocol charges agree. A different exogenous κ_I can change that comparison. This example interprets the proved frontiers; it does not estimate the monetary value of certification or customers' tolerance for risk.

References

- Aardal K, van den Berg P, Gijswijt D, Li S (2014) Approximation algorithms for hard capacitated k -facility location problems. Author manuscript, arXiv:1311.4759v4, September 12.
- Aggarwal A, Klawe MM, Moran S, Shor P, Wilber R (1987) Geometric applications of a matrix-searching algorithm. *Algorithmica* 2:195–208. doi:10.1007/BF01840359.
- Chan TM (2021) (Near-)linear-time randomized algorithms for row minima in Monge partial matrices and related problems. *Proceedings of the 2021 ACM–SIAM Symposium on Discrete Algorithms*, 1465–1482. doi:10.1137/1.9781611976465.88.
- Croci M, Fasi M, Higham NJ, Mary T, Mikaitis M (2022) Stochastic rounding: Implementation, error analysis and applications. *Royal Society Open Science* 9:211631. doi:10.1098/rsos.211631.
- Del Pia A, Dey SS, Molinaro M (2017) Mixed-integer quadratic programming is in NP. *Mathematical Programming* 162:225–240. doi:10.1007/s10107-016-1036-0.
- Eppstein D (2005) PADS: Python Algorithms and Data Structures. *SMAWK.py*, offline matrix-search implementation, originally written 2002 and revised 2005. University of California, Irvine. MIT-licensed distribution; source version and excerpt are identified in the code supplement.
- Fattahi A (2025) Utilities’ managed home charging programs for electric vehicles. *Management Science* 71(12):10205–10224. doi:10.1287/mnsc.2023.00850.
- Fu M (2024) A tutorial on quantized feedback control. *IEEE/CAA Journal of Automatica Sinica* 11(1):5–17. doi:10.1109/JAS.2023.123972.
- Grønlund A, Larsen KG, Mathiasen A, Nielsen JS, Schneider S, Song M (2017) Fast exact k -means, k -medians and Bregman divergence clustering in 1D. Preprint, arXiv:1701.07204, version 4.
- Handler GY, Zang I (1980) A dual algorithm for the constrained shortest path problem. *Networks* 10(4):293–309. doi:10.1002/net.3230100403.
- Huangfu Q, Hall JAJ (2018) Parallelizing the dual revised simplex method. *Mathematical Programming Computation* 10(1):119–142. doi:10.1007/s12532-017-0130-5.
- Hübner T, Gupte A, Rebennack S (2025) Spatial branch-and-bound for nonconvex separable piecewise linear optimization. *INFORMS Journal on Computing*, published online June 27. doi:10.1287/ijoc.2024.0755.
- Jourdain B, Pagès G (2021) Optimal dual quantizers of 1D log-concave distributions: Uniqueness and Lloyd like algorithm. *Journal of Approximation Theory* 267:105581. doi:10.1016/j.jat.2021.105581.
- Kaplan H, Mozes S, Nussbaum Y, Sharir M (2017) Submatrix maximum queries in Monge matrices and partial Monge matrices, and their applications. *ACM Transactions on Algorithms* 13(2), Article 26. doi:10.1145/3039873.

- Klawe MM, Kleitman DJ (1990) An almost linear time algorithm for generalized matrix searching. *SIAM Journal on Discrete Mathematics* 3(1):81–97. doi:10.1137/0403009.
- Lawler EL, Wood DE (1966) Branch-and-bound methods: A survey. *Operations Research* 14(4):699–719. doi:10.1287/opre.14.4.699.
- Lemaréchal C (2001) Lagrangian relaxation. Jünger M, Naddef D, eds. *Computational Combinatorial Optimization*, Lecture Notes in Computer Science 2241 (Springer), 112–156. doi:10.1007/3-540-45586-8_4.
- Li Y, Bertsekas D (2025) An error bound for aggregation in approximate dynamic programming. Author manuscript, arXiv:2507.01324v1, July 2.
- Liao W, Wu J (2015) Matrix inequalities for the difference between arithmetic mean and harmonic mean. *arXiv:1501.04823*.
- Nielsen F, Nock R (2014) Optimal interval clustering: Application to Bregman clustering and statistical mixture learning. Author manuscript, arXiv:1403.2485.
- Pagès G, Wilbertz B (2012) Intrinsic stationarity for vector quantization: Foundation of dual quantization. Author manuscript, arXiv:1010.4642v2, March 27. Original preprint 2010.
- Spear SE, Srivastava S (1987) On repeated moral hazard with discounting. *Review of Economic Studies* 54(4):599–617. doi:10.2307/2297484.
- Stouffer K, Pease M, Tang C, Zimmerman T, Pillitteri V, Lightman S, Hahn A, Saravia S, Sherule A, Thompson M (2023) Guide to operational technology (OT) security. NIST Special Publication 800-82 Rev. 3. doi:10.6028/NIST.SP.800-82r3.
- Thomas J, Worrall T (1988) Self-enforcing wage contracts. *Review of Economic Studies* 55(4):541–554. doi:10.2307/2297404.
- Virtanen P, Gommers R, Oliphant TE, et al. (2020) SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods* 17:261–272. doi:10.1038/s41592-019-0686-2.
- Wu X (1991) Optimal quantization by matrix searching. *Journal of Algorithms* 12(4):663–673. doi:10.1016/0196-6774(91)90039-2.
- Yao FF (1980) Efficient dynamic programming using quadrangle inequalities. *Proceedings of the Twelfth Annual ACM Symposium on Theory of Computing*, 429–435. doi:10.1145/800141.804691.
- Zhang H (2012) Solving an infinite horizon adverse selection model through finite policy graphs. *Operations Research* 60(4):850–864. doi:10.1287/opre.1120.1056.
- Zuse Institute Berlin (2026) PySCIPOpt documentation: Non-linear expressions. Official documentation, accessed September 24. <https://pyscipopt.readthedocs.io/en/latest/tutorials/expressions.html>.

Table EC.3: Individual joint certificates. The first column is the reproducible instance identifier.

Instance	Types	Nodes	Seconds	Lower	Upper	Gap/scale
h2-9-0-0	2	1	0.020	0.79453	0.79511	0.00010
h2-9-0-1	2	1	0.014	0.57578	0.57578	0.00000
h2-9-1-0	2	1	0.018	0.82008	0.82053	0.00007
h2-9-1-1	2	1	0.018	0.83589	0.83595	0.00001
h2-17-0-0	2	1	0.078	0.66865	0.66875	0.00002
h2-17-0-1	2	1	0.042	0.65984	0.65984	0.00000
h2-17-1-0	2	1	0.052	0.71419	0.71429	0.00002
h2-17-1-1	2	1	0.063	1.11961	1.11961	0.00000
h4-9-0-0	4	1	0.030	0.81641	0.81663	0.00004
h4-9-0-1	4	1	0.025	0.64713	0.64713	0.00000
h4-9-1-0	4	1	0.027	0.50484	0.50484	0.00000
h4-9-1-1	4	1	0.030	0.42283	0.42283	0.00000
h4-17-0-0	4	1	0.077	0.49392	0.49412	0.00003
h4-17-0-1	4	1	0.094	0.58137	0.58219	0.00014
h4-17-1-0	4	99	1.943	0.56328	0.56422	0.00015
h4-17-1-1	4	1	0.092	0.65512	0.65535	0.00004
h8-9-0-0	8	1	0.046	0.60338	0.60339	0.00000
h8-9-0-1	8	7	0.160	0.81156	0.81170	0.00002
h8-9-1-0	7	301	4.437	0.64127	0.65375	0.00203
h8-9-1-1	8	1	0.051	0.66771	0.66771	0.00000
h8-17-0-0	8	169	8.047	0.65177	0.65394	0.00036
h8-17-0-1	8	1	0.156	0.62026	0.62116	0.00015
h8-17-1-0	8	201	8.052	0.43832	0.44505	0.00109
h8-17-1-1	8	1	0.158	0.88753	0.88770	0.00003
s4-33-32-0	4	1	0.276	0.57796	0.57874	0.00013
s4-33-32-1	4	1	0.331	0.67633	0.67635	0.00000
s4-33-64-0	4	1	0.274	0.57796	0.57874	0.00013
s4-33-64-1	4	1	0.337	0.67633	0.67635	0.00000
s8-33-16-0	8	1	0.466	0.63893	0.63894	0.00000
s8-33-16-1	8	43	8.306	0.76754	0.76983	0.00037
s16-33-1-0	16	1	0.867	0.54159	0.54233	0.00012
s16-33-1-1	16	1	0.833	0.74255	0.74315	0.00010

Table EC.1: Coarse-relaxation stress cases. The root obligation equals total original capacity. Each initial two-group solve is exact for its reduced model but does not certify the original tolerance.

Case	Commands	Final curvature	Original optimum	Initial lower	Initial gap	Refined gap
0	2	2.5	0.86781	0.86781	0.04219	0.00000
1	2	6.0	0.86562	0.86562	0.04437	0.00000
2	1	2.5	-0.45200	-0.45200	0.11000	0.00000
3	1	6.0	-1.06200	-1.06200	0.67000	0.00000

Caps are $(1/5, 2/5, 4/5)$, weights $(1/5, 3/10, 1/2)$, and ceilings are all one. The first two curvatures are one and two; the third is shown. The catalog is $\{0, 1/4, 1/2, 3/4, 1\}$. All charges are zero except case 3, which charges $1/20$ per installed level. The initial groups are $\{1\}$ and $\{2, 3\}$, with the smallest-cap guard intact. Splitting the latter group gives three singleton histories and zero global gap. Every stage has a checked original-space policy and upper certificate.

Table EC.2: Independent original-model mixed-integer comparisons on every primary case. Times and node counts are tangent/secant pairs. All widths are numerical, not rational certificates.

Case	Seconds	Nodes	Max solver gap	Envelope error	Signed width	Status pair
00	0.02/0.01	1/1	0.0e+00	1.0e-04	1.9e-05	0/0
01	0.28/0.23	1/1	0.0e+00	7.7e-05	-6.7e-16	0/0
02	0.02/0.02	1/1	0.0e+00	7.1e-05	1.7e-05	0/0
03	0.02/0.05	1/1	0.0e+00	7.4e-05	1.8e-15	0/0
04	0.02/0.02	1/1	0.0e+00	7.2e-05	2.7e-05	0/0
05	0.07/0.07	1/1	0.0e+00	7.9e-05	-2.2e-16	0/0
06	0.01/0.03	1/1	0.0e+00	7.3e-05	4.3e-06	0/0
07	0.26/0.14	1/1	0.0e+00	7.7e-05	2.2e-16	0/0
08	0.02/0.02	1/1	5.6e-08	5.6e-05	1.3e-05	0/0
09	0.04/0.03	1/1	0.0e+00	8.5e-05	0.0e+00	0/0
10	0.01/0.01	1/1	0.0e+00	8.0e-05	1.1e-16	0/0
11	0.18/0.22	1/1	0.0e+00	6.7e-05	0.0e+00	0/0

Each envelope has 64 segments and a separate two-second solver limit. Status 0 means the numerical solver reported optimal termination; nonzero statuses would remain in the table. All 24 solves report status 0 here. Signed negative widths at floating roundoff scale are deliberately retained. The full formulation, global numerical bounds, primal values, residuals, probabilities, and individual runtimes are in `results/mip.json`. Exhaustive book references agree with these numerical brackets up to the declared 10^{-6} comparison allowance, which is not an exact upper-bound guarantee.

Table EC.4: Independent numerical MIP comparisons for all 24 primary holdout cases. Times and node counts are tangent/secant pairs. Widths are numerical, not rational certificates.

Instance	Seconds	Nodes	Max MIP gap	Envelope	Width	Prefix seconds
h2-9-0-0	0.03/0.02	1/1	0.0e+00	8.4e-05	0.0e+00	0.015
h2-9-0-1	0.01/0.01	1/1	0.0e+00	4.2e-05	0.0e+00	0.027
h2-9-1-0	0.01/0.01	1/1	0.0e+00	8.6e-05	0.0e+00	0.018
h2-9-1-1	0.01/0.00	1/1	0.0e+00	3.1e-05	0.0e+00	0.022
h2-17-0-0	0.02/0.02	1/1	0.0e+00	5.6e-05	0.0e+00	0.042
h2-17-0-1	0.02/0.02	1/1	0.0e+00	2.8e-05	0.0e+00	0.043
h2-17-1-0	0.01/0.00	1/1	0.0e+00	3.3e-05	1.1e-16	0.040
h2-17-1-1	0.01/0.00	1/1	0.0e+00	1.7e-04	2.2e-16	0.056
h4-9-0-0	0.01/0.02	1/1	0.0e+00	4.7e-05	0.0e+00	0.024
h4-9-0-1	0.01/0.01	1/1	0.0e+00	2.1e-05	0.0e+00	0.026
h4-9-1-0	0.01/0.00	1/1	2.2e-16	1.5e-05	1.1e-16	0.022
h4-9-1-1	0.01/0.01	1/1	0.0e+00	1.2e-05	4.8e-06	0.027
h4-17-0-0	0.03/0.02	1/1	0.0e+00	1.7e-05	1.7e-16	0.065
h4-17-0-1	0.03/0.02	1/1	0.0e+00	2.6e-05	1.1e-16	0.120
h4-17-1-0	0.03/0.03	1/1	0.0e+00	3.4e-05	0.0e+00	0.087
h4-17-1-1	0.01/0.01	1/1	0.0e+00	1.8e-05	0.0e+00	0.125
h8-9-0-0	0.02/0.01	1/1	0.0e+00	6.4e-05	8.5e-05	0.041
h8-9-0-1	0.05/0.05	1/1	0.0e+00	3.5e-05	5.3e-15	0.075
h8-9-1-0	0.04/0.04	1/1	0.0e+00	6.3e-05	0.0e+00	0.063
h8-9-1-1	0.01/0.01	1/1	0.0e+00	3.4e-05	4.1e-05	0.045
h8-17-0-0	0.12/0.11	1/1	0.0e+00	5.8e-05	0.0e+00	0.179
h8-17-0-1	0.03/0.03	1/1	0.0e+00	3.6e-05	0.0e+00	0.123
h8-17-1-0	0.01/0.01	1/1	1.3e-16	1.5e-05	5.6e-17	0.170
h8-17-1-1	0.02/0.02	1/1	0.0e+00	4.9e-05	2.2e-06	0.128

Each envelope solve has a four-second limit and 64 shortfall segments. All 48 solves returned optimal status in this execution; hence no timeout is hidden. Maximum mixed-integer primal–dual gaps and the per-case shortfall-envelope errors are reported above. Signed raw widths, including negative values at floating roundoff scale, solver messages, residuals, solutions, and separate node counts are retained in `comparators.json`. Display widths truncate only negative floating residuals to zero; this is not a rational exactness assertion. The companion also retains the separate eight-case initial MIP execution, which is not pooled into the 24-case comparison.

Table EC.5: Accounting for Theorem D.6; ordered rational input, $1 \leq m \leq k$.

Cost component	Included work or storage
Input validation and six prefix moments	$O(k)$ rational operations; $O(k)$ stored values.
Sorting	Input interface requires ordered distinct caps; otherwise add $O(k \log k)$ exact comparisons and reorder associated primitives.
Continuous terminal minimization	Included in $O(k)$ oracle requests; each interval minimizes its quadratic exactly, including singleton and endpoint cases.
Key comparisons and repeated requests	Included in $O(mk)$ work; a repeated finite request is charged again. No quadratic entry cache is assumed.
Padding comparisons	Included index comparisons; undefined economic costs are never evaluated.
Previous-layer values and backpointers	$O(mk)$ stored values and indices, plus $O(k)$ working search space.
Reconstruction and output	One winning codebook per budget; $O(m^2) \leq O(mk)$ symbols written. No reconstruction at every candidate endpoint.
Rational heights and integer operations	$O(S)$ bits per evaluated rational, $S = O(k[L + \log(k+2)])$; conservative $O(mkS^3)$ bit operations, including reduction and comparison.
Physical bit storage	$O((mk + k)[S + \log(k+2)])$ bits for the stated arrays and indices; not a claim of $O(mk)$ bits.
Inherited public graph and response compiler	Not invoked by the renewal algorithm; their read-only data, compilation, and query costs enter the separate implementation menu when used.
External solver preprocessing	Not part of the frontier theorem. Finite-network construction, floating solver work, and exact certification are measured separately.